Block practice · Identities, equations and inequalities

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Drawing the lines for an inequality graph

Factor theorem: show a factor or find a constant

Shading or describing a region

Factorising and solving a cubic

Where a line meets a curve (simultaneous equations)

Quadratic inequalities

Two constants from factor and remainder information

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Practice workbook · Drawing the lines for an inequality graph

Drawing the straight boundary lines of a set of linear inequalities accurately on a grid or axes.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-05

1. Question 5 · part a · 3 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part a only.

5 (a) On the grid opposite draw the line with equation

(i) $y = 2x + 5$ (ii) $4y = x - 8$ (iii) $5y + 3x = 30$

(3)

(b) Show, by shading, the region R defined by the inequalities

$y \leqslant 2x + 5$ $4y \geqslant x - 8$ $5y + 3x \leqslant 30$

(1)

For all points in R with coordinates (x, y)

$P = 2x - 5y$

(c) Using your graph, find the least value of P

(3)

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1. Question 5 · part a · 3 marks · Working and answer

Working and answer

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2. Question 2 · part a · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

2 (a) Using the axes below sketch the line with equation

(i) $y = 6$ (ii) $y + x = 10$ (iii) $y = 2x - 5$

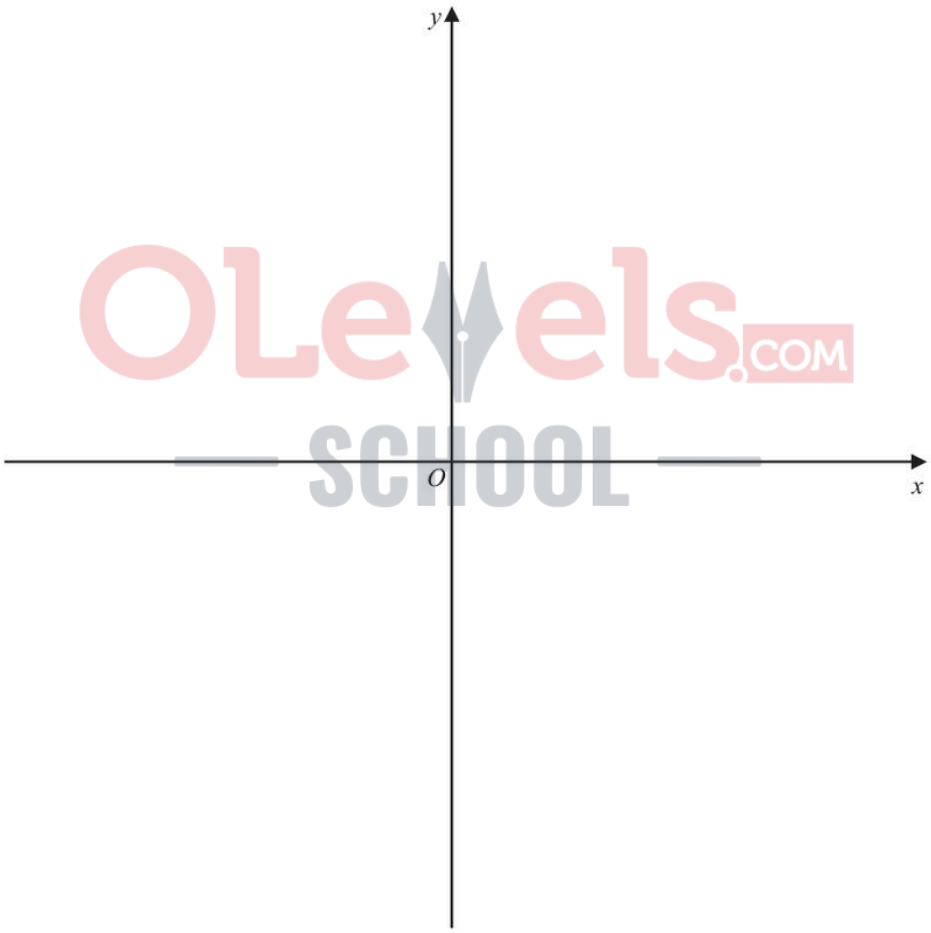
Show the coordinates of any point where each line crosses the coordinate axes.

(3)

(b) Show, by shading on your sketch, the region R defined by the inequalities

$y \leqslant 6$ $y + x \leqslant 10$ $y \geqslant 2x - 5$ $x \geqslant 0$

(1)



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2. Question 2 · part a · 3 marks · Working and answer

Working and answer

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3. Question 4 · part a.i · 3 marks

2022 JAN · Paper 01

Further practice · Attempt part a.i only.

4 (a) On the grid opposite, draw the line with equation

(i) $y = 2x - 4$ (ii) $2x + 3y = 12$ (iii) $y + 2x + 2 = 0$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$y \geqslant 2x - 4$ $2x + 3y \leqslant 12$ $y + 2x + 2 \geqslant 0$

(1)

For all points in R, with coordinates (x, y)

$$P = x - 2y$$

(c) find the least value of P

(4)

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3. Question 4 · part a.i · 3 marks · Working and answer

Working and answer

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4. Question 4 · part a · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

4 (a) On the axes opposite, draw the line with equation

(i) $y = -x - 1$ (ii) $y - 3x + 8 = 0$ (iii) $2y = x + 8$

(3)

(b) Show, by shading on your graph, the region R defined by the inequalities

$y \geqslant -x - 1$ and $y \geqslant 3x - 8$ and $2y \leqslant x + 8$

(1)

For all points in R, with coordinates (x, y)

$$P = 2y - 3x$$

(c) Find

(i) the greatest value of P

(ii) the least value of P

(4)

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P 7 3 5 8 6 A 0 8 3 2

4. Question 4 · part a · 3 marks · Working and answer

Working and answer

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5. Question 4 · part a · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

4 (a) On the grid opposite, draw the graph of the line with equation

(i) $3x+2y=18$ (ii) $x+3y-6=0$ (iii) $y=3x$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$3x+2y \leqslant 18$ $x+3y-6 \geqslant 0$ $y \leqslant 3x$

(1)

For all points in R, with coordinates (x, y)

$$P=2x-y$$

(c) find the least value of P

(4)

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5. Question 4 · part a · 3 marks · Working and answer

Working and answer

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6. Question 5 · part a · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

5 (a) On the grid opposite, draw the graphs of the lines with equations

$$2x + 3y = 24 \qquad y = 2x \qquad 3y = 2x - 12$$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$$2x + 3y \leqslant 24 \qquad y \leqslant 2x \qquad 3y \geqslant 2x - 12 \qquad y \geqslant 0$$

(1)

For all points in R, with coordinates (x, y)

$$F = 2x + 5y$$

(c) Find the greatest value of F.

(3)

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6. Question 5 · part a · 3 marks · Working and answer

Working and answer

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7. Question 2 · part a · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part a only.

2 (a) On the grid below, draw the line with equation

(i) $5x + 2y = 10$ (ii) $y = x$

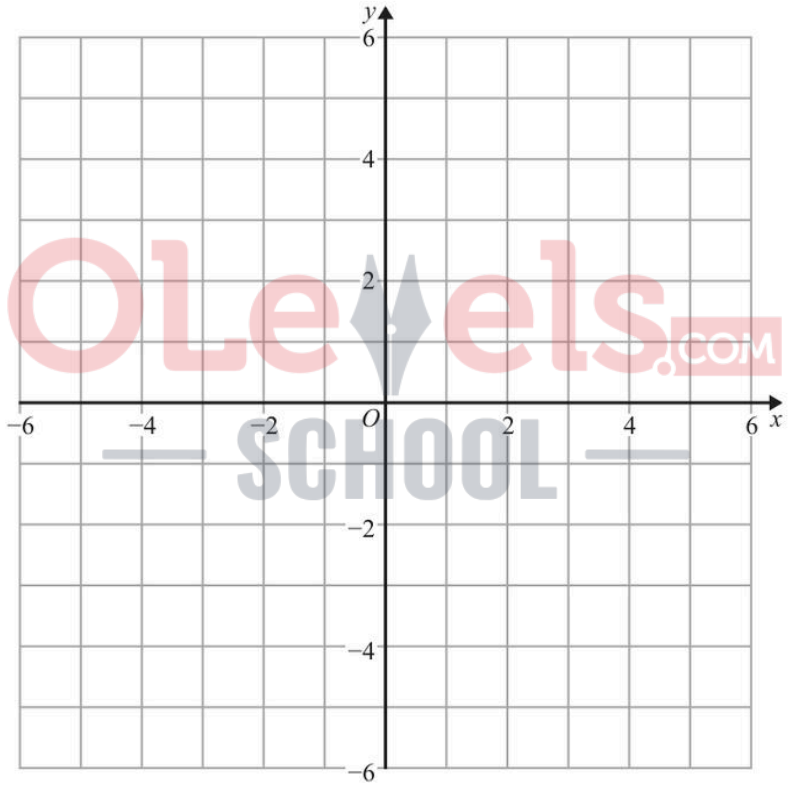
(2)

(b) Show, by shading on the grid, the region R defined by the inequalities

$y \leqslant x$ $5x + 2y \leqslant 10$ $y \geqslant -2$ $x \geqslant 1$

Label the region R.

(2)



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6

P 5 9 9 3 8 A 0 6 3 6

7. Question 2 · part a · 2 marks · Working and answer

Working and answer

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8. Question 1 · part a.i · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part a.i only.

1 (a) On the grid below, draw the line with equation

(i) $3x+4y=24$ (ii) $2x-5y+10=0$

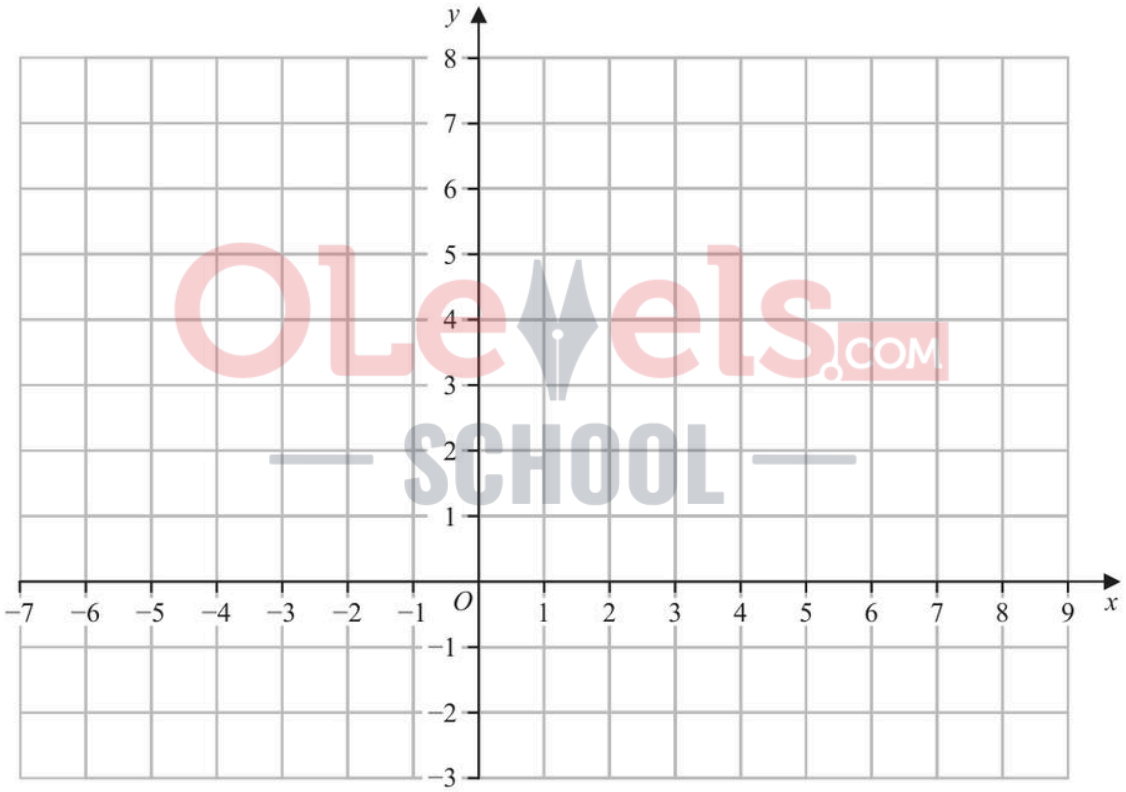
(2)

(b) Show, by shading on the grid, the region R defined by the inequalities

$3x+4y \leqslant 24$ $2x-5y+10 \geqslant 0$ $y \leqslant 5$ $x \geqslant -1$

Label the region R

(2)



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8. Question 1 · part a.i · 2 marks · Working and answer

Working and answer

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9. Question 1 · part a · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

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1 (a) On the grid below, draw the graph of the line with equation

(i) $4x + 5y = 20$ (ii) $3y - 4x = -12$

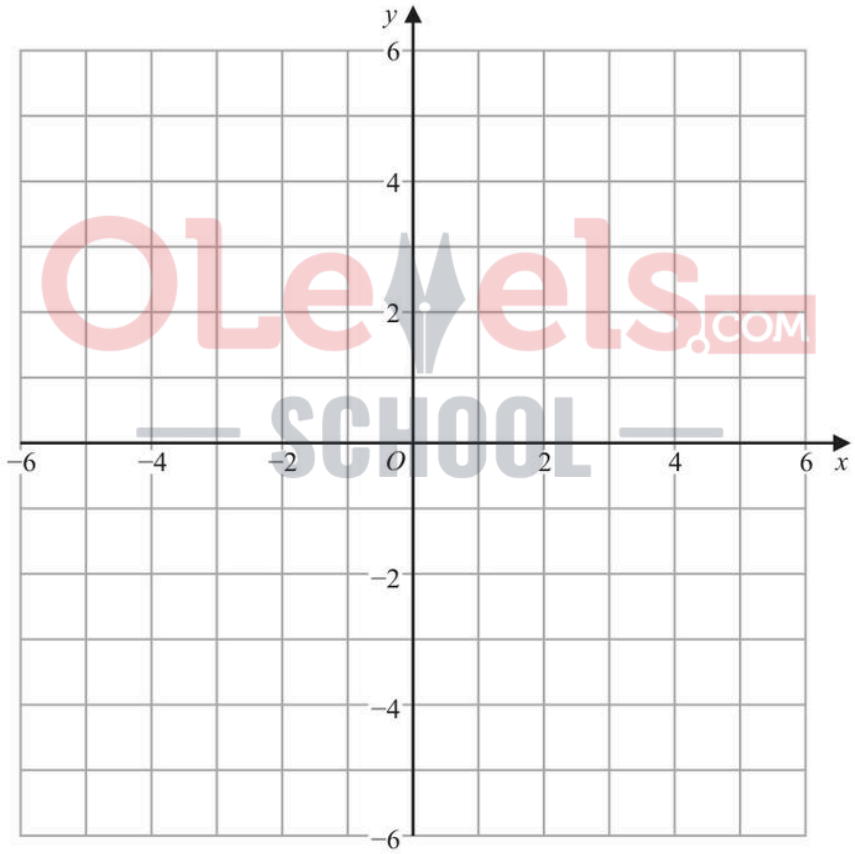
(2)

(b) Show, by shading on the grid, the region R defined by the inequalities

$4x + 5y \leqslant 20$ $3y - 4x \geqslant -12$ $y \leqslant 3$ $x \geqslant 1$

Label the region R.

(2)



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Turn over for a spare grid if you need to redraw your graph.

9. Question 1 · part a · 2 marks · Working and answer

Working and answer

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10. Question 5 · part a.i · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part a.i only.

5 (a) On the grid opposite, draw the line with equation

(i) $3x-4y=12$ (ii) $y+6+3x=0$ (iii) $3y=18-x$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$3x-4y \leqslant 12$ $y+6+3x \geqslant 0$ $3y \leqslant 18-x$

(1)

For all points in R, with coordinates (x, y)

$$P=3x-2y$$

Using values from your graph,

(c) find the least value of P and the greatest value of P

(4)

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P 7 8 9 6 4 A 0 1 2 3 6

10. Question 5 · part a.i · 3 marks · Working and answer

Working and answer

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Practice workbook · Factor theorem: show a factor or find a constant

Substituting a single value into a polynomial to show that a given linear expression is a factor, to find one unknown constant, or to produce a relation between constants.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-08

1. Question 3 · part a · 2 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part a only.

3 Given that $(x - 4)$ is a factor of $px^3 - 31x^2 + 25x + 12$ where p is a constant,

(a) show that $p = 6$

(2)

(b) Solve the equation $6x^3 - 31x^2 + 25x + 12 = 0$

Show clear algebraic working.

(4)

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1. Question 3 · part a · 2 marks · Working and answer

Working and answer

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2. Question 9 · part b.i · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

9

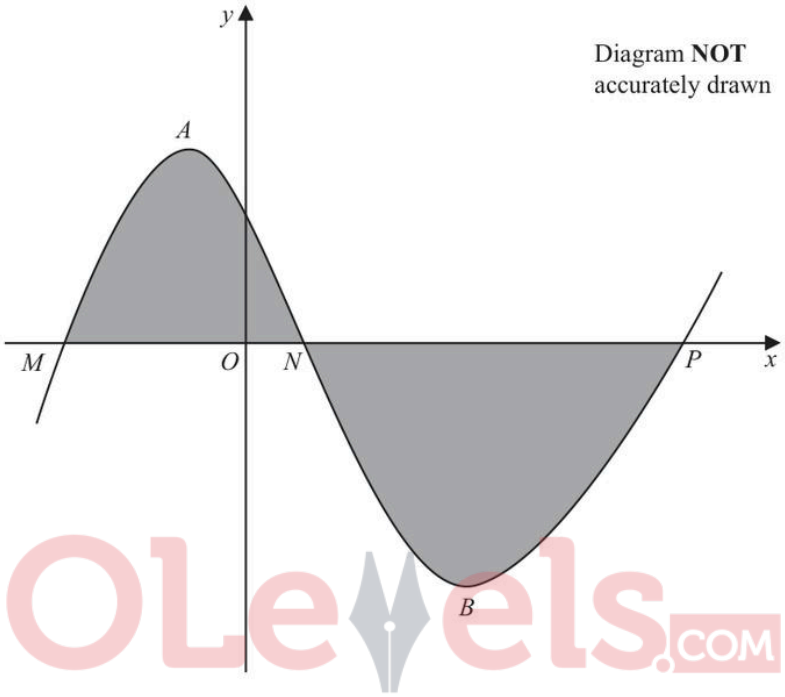


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^3 + ax^2 + bx + c$$

The curve C has a maximum at the point A with coordinates $\left(-\frac{1}{3}, \frac{100}{27}\right)$ and a minimum at the point B with coordinates $(2, -9)$

Given that a, b and c are integers

(a) show that $a = -5, b = -4$ and $c = 3$

(5)

(b) (i) Show that $(x+1)$ is a factor of $f(x)$

(1)

(ii) Hence, or otherwise, use algebra to factorise $f(x)$ completely.

(3)

The curve C crosses the x-axis at the points M, N and P
The finite regions shown shaded in Figure 4 are bounded by the curve C and parts of the x-axis from M to N and from N to P

(c) Use algebraic integration to determine the total area of the shaded regions. Give your answer as an exact fraction.

(4)

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P 7 2 8 6 5 A 0 2 8 3 6

2. Question 9 · part b.i · 1 mark · Working and answer

Working and answer

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3. Question 1 · part a · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

1 $f(x) = (x - 3)[x^2 + (p - 2)x + q]$

Given that $f(0) = -12$

(a) find the value of q.

(2)

(b) Find the range of values of p for which the cubic equation $f(x) = 0$ has only one real root.

(5)

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(Total for Question 1 is 7 marks)

3. Question 1 · part a · 2 marks · Working and answer

Working and answer

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4. Question 2 · part b · 1 mark

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

2 $f(x) = x^3 + px + q$ where p and q are constants.

The remainder when $f(x)$ is divided by $(x - 1)$ is -12

The remainder when $f(x)$ is divided by $(x - 4)$ is 30

(a) Find the value of p and the value of q.

(6)

Using your values of p and q

(b) show that $f(3) = 0$

(1)

(c) Express $f(x)$ as a product of linear factors.

(3)

(d) Hence solve the equation $f(x) = 0$

(1)

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4. Question 2 · part b · 1 mark · Working and answer

Working and answer

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5. Question 10 · part a · 1 mark

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part a only.

10 $f(x) = 32x^3 - 33x + 1$

(a) Show that $f(1) = 0$

(1)

(b) Hence using an algebraic method solve $f(x) = 0$

(4)

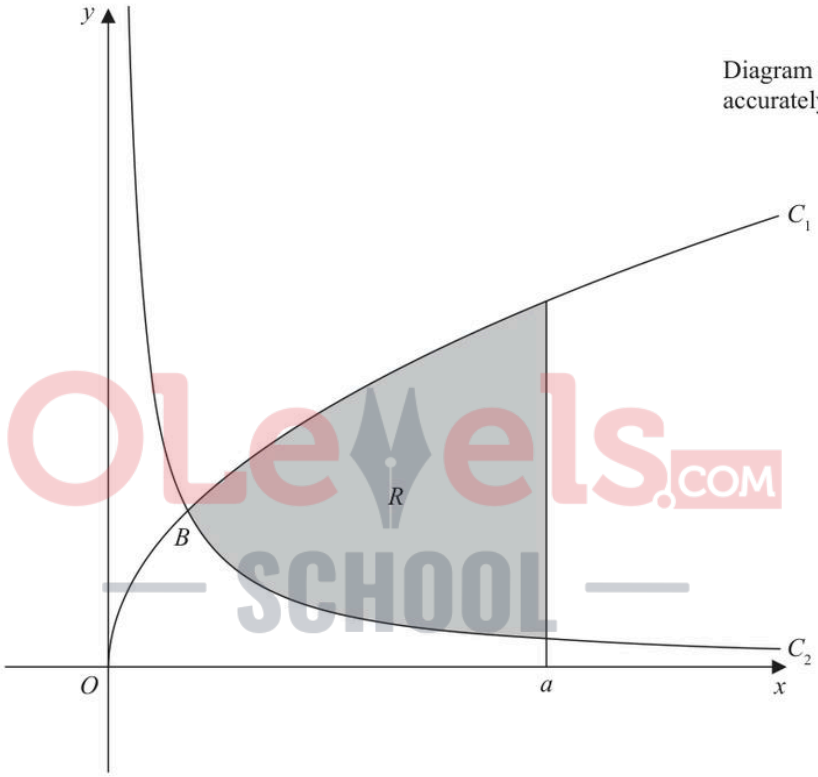


Figure 4

The region R, shown shaded in Figure 4, is bounded by the curve C_1 with equation $y = \sqrt{x}$, by the curve C_2 with equation $y = \frac{1}{8x}$ and by the line with equation $x = a$

The curves C_1 and C_2 intersect at the point B, with x coordinate p, where $p < a$

(c) Find the value of p.

(2)

The region R is rotated through 360° about the x-axis to generate a solid with volume $\frac{27\pi}{64}$

(d) Use algebraic integration to find the value of a.

(7)

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5. Question 10 · part a · 1 mark · Working and answer

Working and answer

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6. Question 6 · part a · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

6 $f(x) = x^3 + (p + 1)x^2 - 10x + q$

where p and q are integers.

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that $9p + q + 6 = 0$

(3)

Given that $(x + p)$, where $p > 0$, is also a factor of $f(x)$

(b) show that $p^2 + 10p + q = 0$

(3)

(c) Hence find the value of p and the value of q.

(5)

(d) Using your values of p and q, factorise $f(x)$ completely.

(2)

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6. Question 6 · part a · 3 marks · Working and answer

Working and answer

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7. Question 6 · part b · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

6 $f(x) = x^3 + (p + 1)x^2 - 10x + q$

where p and q are integers.

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that $9p + q + 6 = 0$

(3)

Given that $(x + p)$, where $p > 0$, is also a factor of $f(x)$

(b) show that $p^2 + 10p + q = 0$

(3)

(c) Hence find the value of p and the value of q.

(5)

(d) Using your values of p and q, factorise $f(x)$ completely.

(2)

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7. Question 6 · part b · 3 marks · Working and answer

Working and answer

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8. Question 1 · part a · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

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1 $f(x) = 6x^3 - 13x^2 + ax - 10$ where a is a constant

Given that $(3x - 2)$ is a factor of $f(x)$

(a) show that $a = 21$

(2)

(b) Hence show algebraically that the curve $y = f(x)$ has only one intersection with the x-axis.

(4)

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(Total for Question 1 is 6 marks)

8. Question 1 · part a · 2 marks · Working and answer

Working and answer

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9. Question 7 · part a · 2 marks

2024 NOV · Paper 02

Further practice · Attempt part a only.

7 (a) Use the factor theorem to show that $(4x-1)$ is a factor of

$$f(x) = 64x^3 - 64x^2 + 3$$

(2)

(b) Hence, or otherwise, find the exact roots of the equation

$$f(x) = 0$$

(4)

A geometric series G has first term a and common ratio r
The third term of G is 9 and the sum to infinity of G is 192

(c) Show that $64r^3 - 64r^2 + 3 = 0$

(3)

Given that r is a rational number

(d) write down the value of r

(1)

(e) show that $a = 144$

(2)

The sum to n terms of G is S_n

(f) Using logarithms, find the least value of n such that $S_n > 191.9$

(4)

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9. Question 7 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 10 · part a · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

10 (a) Use the factor theorem to show that $(4x - 3)$ is a factor of

$$16x^3 + 11x - 15$$

(2)

(b) Using formulae given on page 2, show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

(c) Show that the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

becomes the equation

$$16x^3 + 11x - 15 = 0$$

by using the substitution $x = \cos\theta$

(4)

(d) Hence show that any solution of the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

is given by $\cos\theta = \frac{3}{4}$

(4)

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10. Question 10 · part a · 2 marks · Working and answer

Working and answer

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Practice workbook · Shading or describing a region

Showing the region that satisfies several linear inequalities by shading, or doing the reverse and writing the inequalities that describe a given region.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-06

1. Question 5 · part b · 1 mark

2023 JUN · Paper 02

Worked-example retry · Attempt part b only.

5 (a) On the grid opposite draw the line with equation

(i) $y = 2x + 5$ (ii) $4y = x - 8$ (iii) $5y + 3x = 30$

(3)

(b) Show, by shading, the region R defined by the inequalities

$y \leqslant 2x + 5$ $4y \geqslant x - 8$ $5y + 3x \leqslant 30$

(1)

For all points in R with coordinates (x, y)

$P = 2x - 5y$

(c) Using your graph, find the least value of P

(3)

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1. Question 5 · part b · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 2 · part b · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b only.

2 (a) Using the axes below sketch the line with equation

(i) $y = 6$ (ii) $y + x = 10$ (iii) $y = 2x - 5$

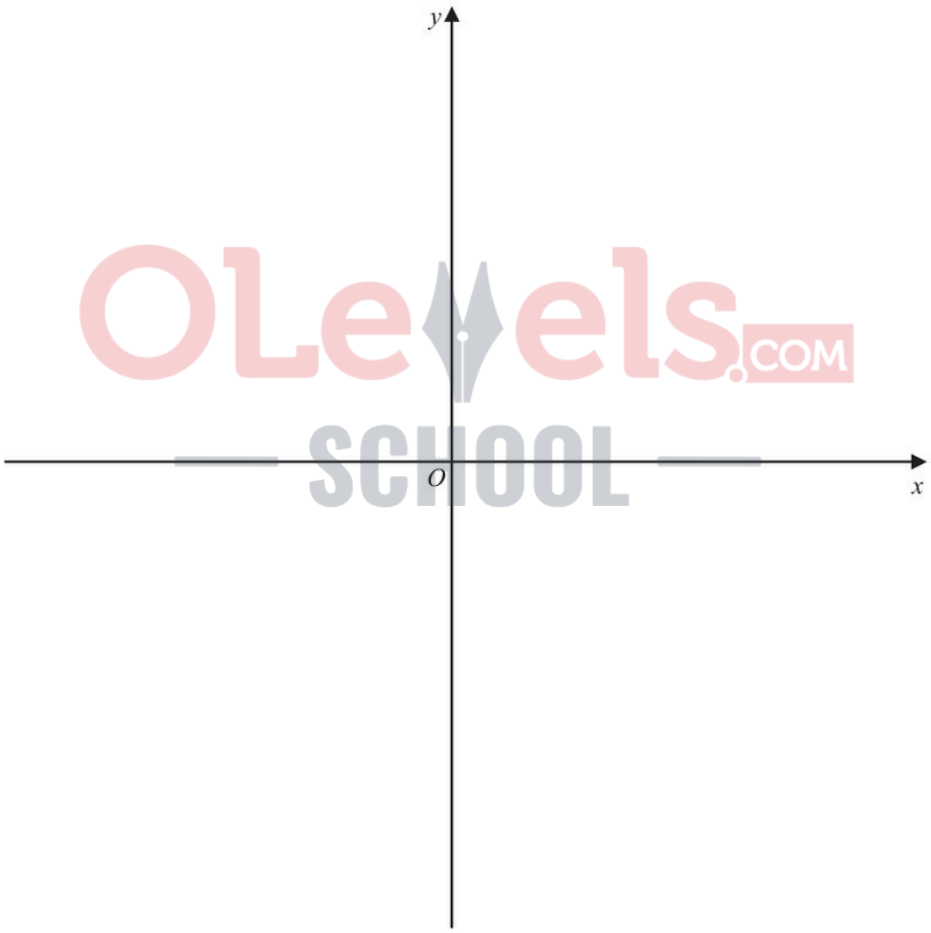
Show the coordinates of any point where each line crosses the coordinate axes.

(3)

(b) Show, by shading on your sketch, the region R defined by the inequalities

$y \leqslant 6$ $y + x \leqslant 10$ $y \geqslant 2x - 5$ $x \geqslant 0$

(1)



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P 6 2 2 8 0 A 0 4 3 2

2. Question 2 · part b · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 4 marks)

2. Question 2 · part b · 1 mark · Working and answer

Working and answer

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3. Question 4 · part b · 1 mark

2022 JAN · Paper 01

Further practice · Attempt part b only.

4 (a) On the grid opposite, draw the line with equation

(i) $y = 2x - 4$ (ii) $2x + 3y = 12$ (iii) $y + 2x + 2 = 0$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$y \geqslant 2x - 4$ $2x + 3y \leqslant 12$ $y + 2x + 2 \geqslant 0$

(1)

For all points in R, with coordinates (x, y)

$$P = x - 2y$$

(c) find the least value of P

(4)

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3. Question 4 · part b · 1 mark · Working and answer

Working and answer

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4. Question 4 · part b · 1 mark

2023 NOV · Paper 02

Further practice · Attempt part b only.

4 (a) On the axes opposite, draw the line with equation

(i) $y = -x - 1$ (ii) $y - 3x + 8 = 0$ (iii) $2y = x + 8$

(3)

(b) Show, by shading on your graph, the region R defined by the inequalities

$y \geqslant -x - 1$ and $y \geqslant 3x - 8$ and $2y \leqslant x + 8$

(1)

For all points in R, with coordinates (x, y)

$$P = 2y - 3x$$

(c) Find

(i) the greatest value of P

(ii) the least value of P

(4)

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P 7 3 5 8 6 A 0 8 3 2

4. Question 4 · part b · 1 mark · Working and answer

Working and answer

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5. Question 4 · part b · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b only.

4 (a) On the grid opposite, draw the graph of the line with equation

(i) $3x+2y=18$ (ii) $x+3y-6=0$ (iii) $y=3x$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$3x+2y \leqslant 18$ $x+3y-6 \geqslant 0$ $y \leqslant 3x$

(1)

For all points in R, with coordinates (x, y)

$$P=2x-y$$

(c) find the least value of P

(4)

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5. Question 4 · part b · 1 mark · Working and answer

Working and answer

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6. Question 5 · part b · 1 mark

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

5 (a) On the grid opposite, draw the graphs of the lines with equations

$$2x + 3y = 24 \qquad y = 2x \qquad 3y = 2x - 12$$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$$2x + 3y \leqslant 24 \qquad y \leqslant 2x \qquad 3y \geqslant 2x - 12 \qquad y \geqslant 0$$

(1)

For all points in R, with coordinates (x, y)

$$F = 2x + 5y$$

(c) Find the greatest value of F.

(3)

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6. Question 5 · part b · 1 mark · Working and answer

Working and answer

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7. Question 5 · part b · 1 mark

2025 NOV · Paper 01

Further practice · Attempt part b only.

5 (a) On the grid opposite, draw the line with equation

(i) $3x-4y=12$ (ii) $y+6+3x=0$ (iii) $3y=18-x$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$3x-4y \leqslant 12$ $y+6+3x \geqslant 0$ $3y \leqslant 18-x$

(1)

For all points in R, with coordinates (x, y)

$$P=3x-2y$$

Using values from your graph,

(c) find the least value of P and the greatest value of P

(4)

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7. Question 5 · part b · 1 mark · Working and answer

Working and answer

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8. Question 2 · part b · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part b only.

2 (a) On the grid below, draw the line with equation

(i) $5x + 2y = 10$ (ii) $y = x$

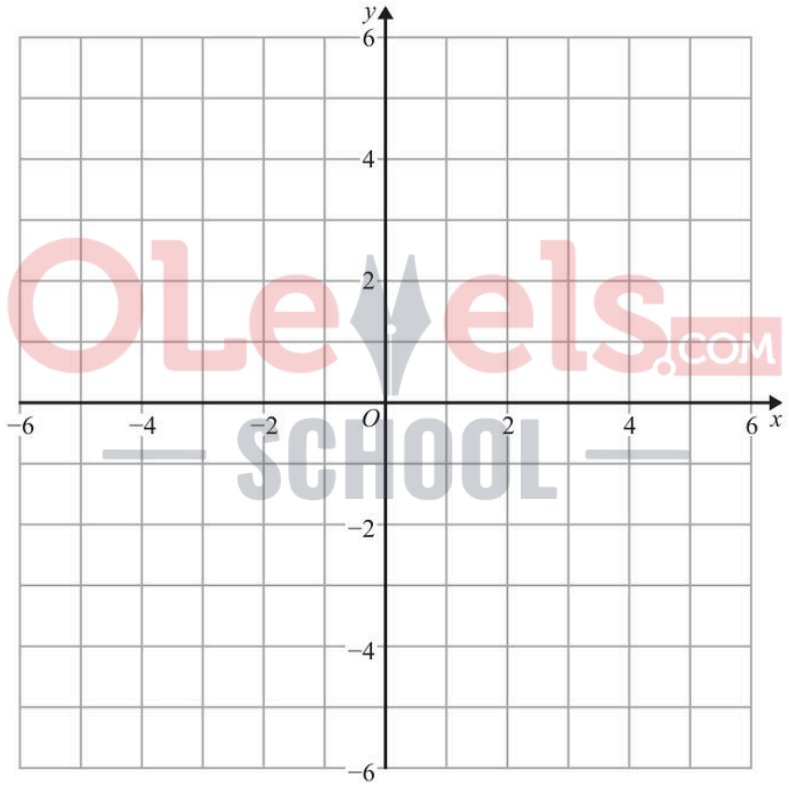
(2)

(b) Show, by shading on the grid, the region R defined by the inequalities

$y \leqslant x$ $5x + 2y \leqslant 10$ $y \geqslant -2$ $x \geqslant 1$

Label the region R.

(2)



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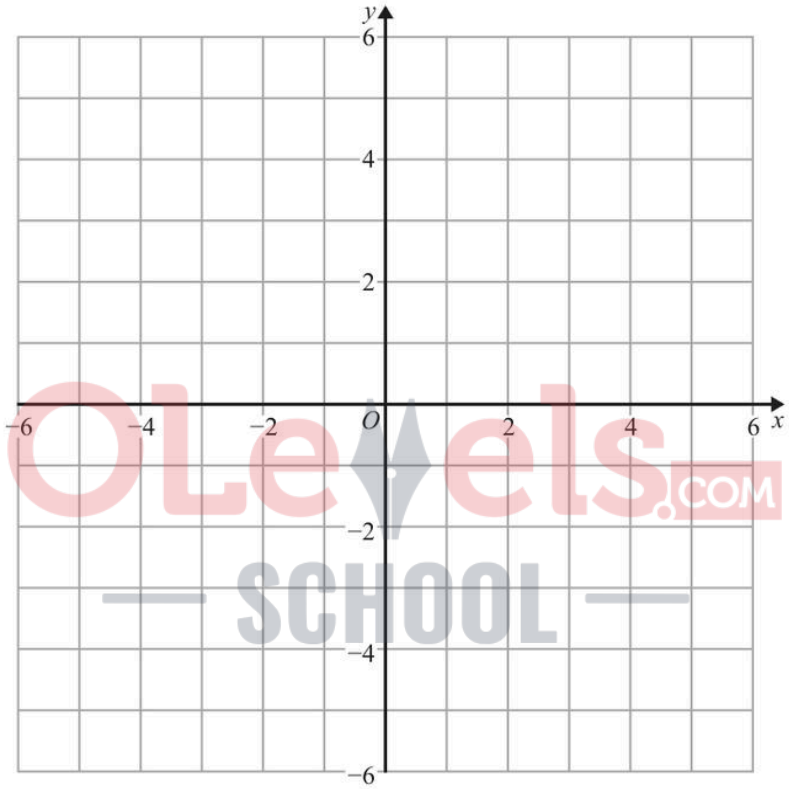
8. Question 2 · part b · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part b only.

Question 2 continued

Only use this grid if you need to redraw your graph.



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(Total for Question 2 is 4 marks)

8. Question 2 · part b · 2 marks · Working and answer

Working and answer

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9. Question 1 · part b · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

1 (a) On the grid below, draw the line with equation

(i) $3x+4y=24$ (ii) $2x-5y+10=0$

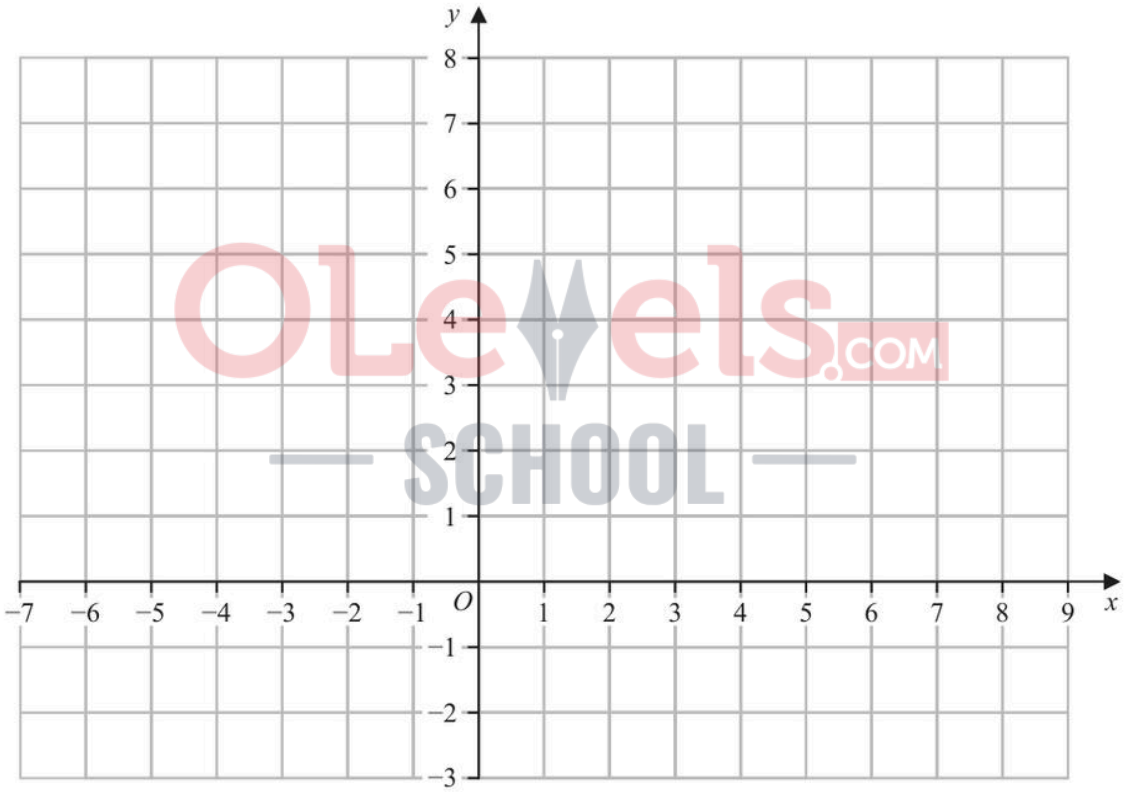
(2)

(b) Show, by shading on the grid, the region R defined by the inequalities

$3x+4y \leqslant 24$ $2x-5y+10 \geqslant 0$ $y \leqslant 5$ $x \geqslant -1$

Label the region R

(2)



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9. Question 1 · part b · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

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Question 1 continued

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(Total for Question 1 is 4 marks)

9. Question 1 · part b · 2 marks · Working and answer

Working and answer

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10. Question 1 · part b · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

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1 (a) On the grid below, draw the graph of the line with equation

(i) $4x + 5y = 20$ (ii) $3y - 4x = -12$

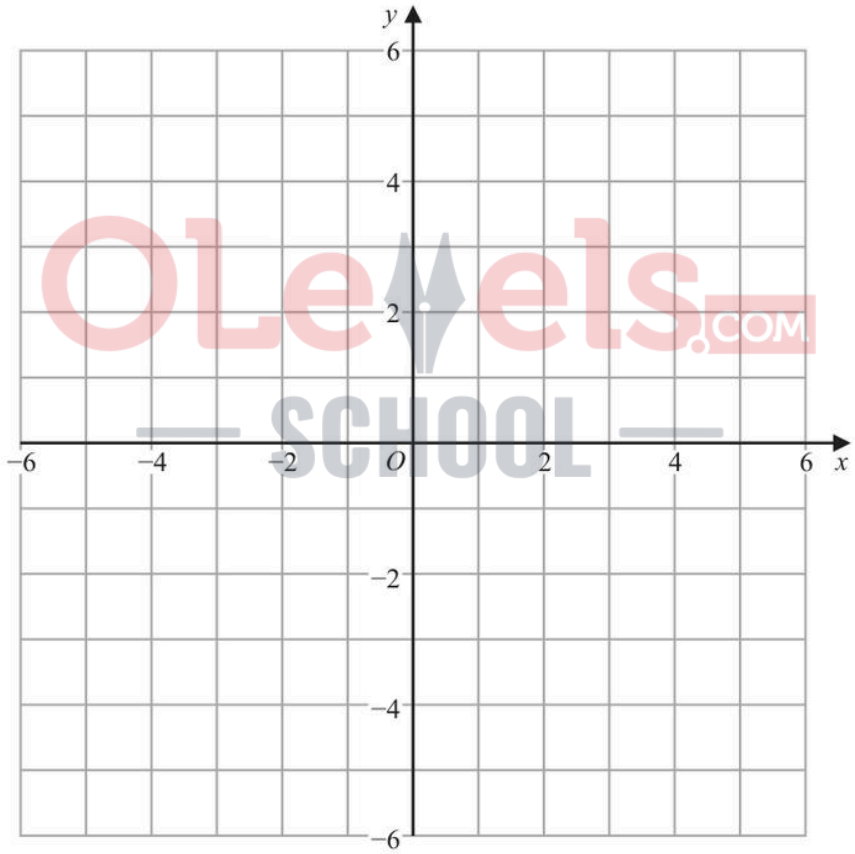
(2)

(b) Show, by shading on the grid, the region R defined by the inequalities

$4x + 5y \leqslant 20$ $3y - 4x \geqslant -12$ $y \leqslant 3$ $x \geqslant 1$

Label the region R.

(2)



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Turn over for a spare grid if you need to redraw your graph.

10. Question 1 · part b · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 1 continued

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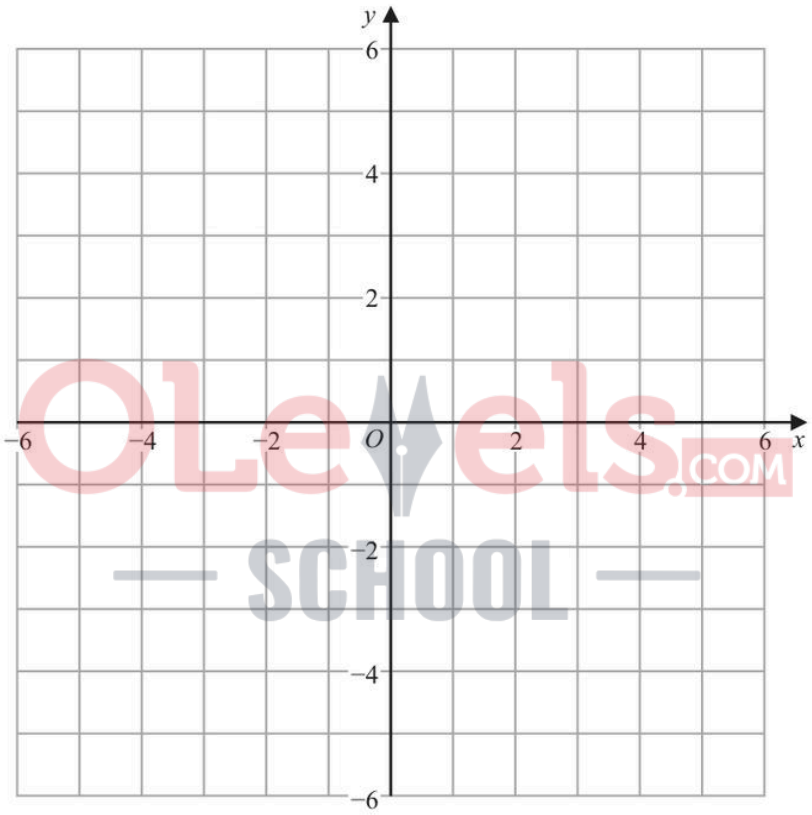
10. Question 1 · part b · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 1 continued

Only use this grid if you need to redraw your graph.



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(Total for Question 1 is 4 marks)

10. Question 1 · part b · 2 marks · Working and answer

Working and answer

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11. Question 2 · part b · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

2

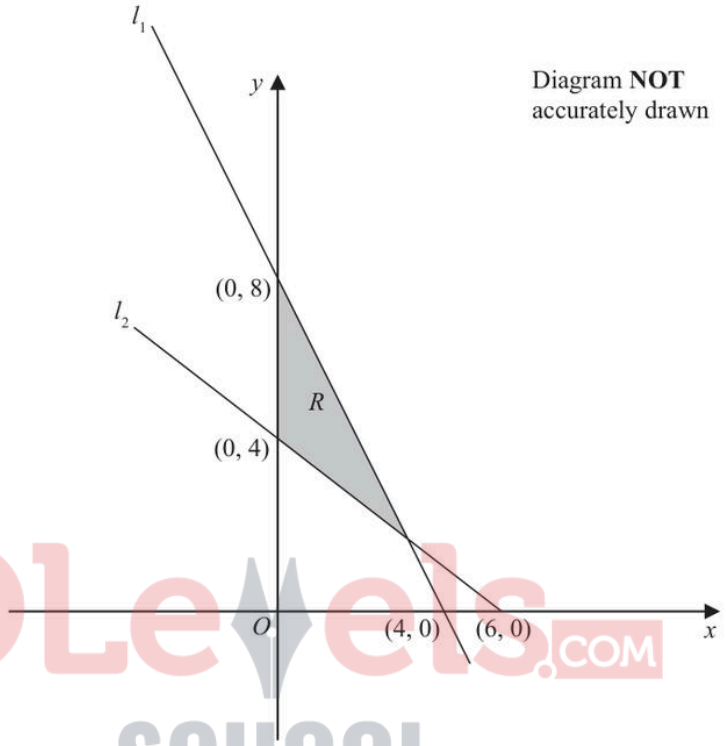


Figure 2

Figure 2 shows the shaded region R bounded by the line l_1, the line l_2 and the y-axis.

The points with coordinates (0, 8) and (4, 0) lie on l_1

The points with coordinates (0, 4) and (6, 0) lie on l_2

(a) Find, in the form $ax + by = c$, where a, b and c integers, an equation of

(i) l_1

(ii) l_2

(3)

(b) Hence write down three inequalities that define the region R.

(3)

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11. Question 2 · part b · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 6 marks)

11. Question 2 · part b · 3 marks · Working and answer

Working and answer

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Practice workbook · Factorising and solving a cubic

Dividing a cubic by a known linear factor and factorising the quadratic quotient, to write f(x) as a product of factors or to solve f(x) = 0.

22 question parts: 1 worked-example retry and 21 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-10

1. Question 3 · part b · 4 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

3 Given that $(x - 4)$ is a factor of $px^3 - 31x^2 + 25x + 12$ where p is a constant,

(a) show that $p = 6$

(2)

(b) Solve the equation $6x^3 - 31x^2 + 25x + 12 = 0$

Show clear algebraic working.

(4)

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1. Question 3 · part b · 4 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 6 marks)

1. Question 3 · part b · 4 marks · Working and answer

Working and answer

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2. Question 9 · part b.ii · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part b.ii only.

9

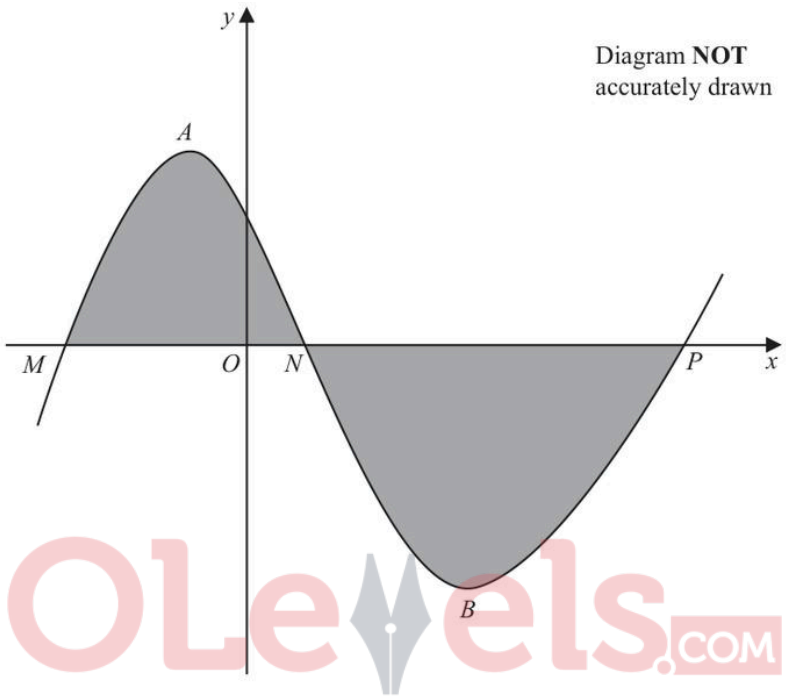


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^3 + ax^2 + bx + c$$

The curve C has a maximum at the point A with coordinates $\left(-\frac{1}{3}, \frac{100}{27}\right)$ and a minimum at the point B with coordinates $(2, -9)$

Given that a, b and c are integers

(a) show that $a = -5, b = -4$ and $c = 3$

(5)

(b) (i) Show that $(x+1)$ is a factor of $f(x)$

(1)

(ii) Hence, or otherwise, use algebra to factorise $f(x)$ completely.

(3)

The curve C crosses the x-axis at the points M, N and P
The finite regions shown shaded in Figure 4 are bounded by the curve C and parts of the x-axis from M to N and from N to P

(c) Use algebraic integration to determine the total area of the shaded regions. Give your answer as an exact fraction.

(4)

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2. Question 9 · part b.ii · 3 marks · Working and answer

Working and answer

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3. Question 2 · part c · 3 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part c only.

2 $f(x) = x^3 + px + q$ where p and q are constants.

The remainder when $f(x)$ is divided by $(x - 1)$ is -12

The remainder when $f(x)$ is divided by $(x - 4)$ is 30

(a) Find the value of p and the value of q.

(6)

Using your values of p and q

(b) show that $f(3) = 0$

(1)

(c) Express $f(x)$ as a product of linear factors.

(3)

(d) Hence solve the equation $f(x) = 0$

(1)

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3. Question 2 · part c · 3 marks · Working and answer

Working and answer

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4. Question 4 · part b · 3 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

4 $f(x) = 2x^3 + ax^2 + bx + 18$ where a and b are constants

When $f'(x)$ is divided by $(x - 2)$ the remainder is 5

Given that $(x - 2)$ is a factor of $f(x)$

(a) find the value of a and the value of b.

(6)

(b) Express $f(x)$ as a product of linear factors.

(3)

(c) Hence use algebra to solve the equation $f(x) = 0$

(2)

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4. Question 4 · part b · 3 marks · Working and answer

Working and answer

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5. Question 5 · part b · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

5 $f(x) = 2x^3 + ax^2 - 14x + b$ where a and b are constants.

When $f(x)$ is divided by $(x - 4)$ the remainder is 39

Given that $(x - 1)$ is a factor of $f(x)$

(a) show that $a = -3$ and find the value of b

(5)

(b) Hence factorise $f(x)$ completely.

(4)

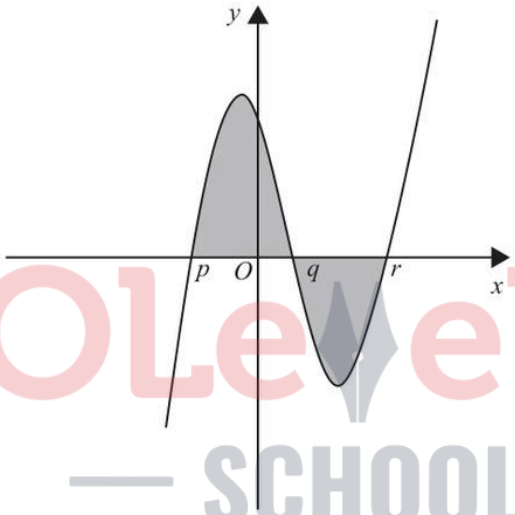


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = f(x)$

Given that C crosses the x-axis at the points with coordinates $(p, 0)$, $(q, 0)$ and $(r, 0)$

(c) write down the value of p, the value of q and the value of r

(3)

The region shown shaded in Figure 3 is bounded by the curve and the x-axis.

(d) Use algebraic integration to find the exact area of the shaded region.

(4)

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5. Question 5 · part b · 4 marks · Working and answer

Working and answer

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6. Question 4 · part b · 3 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

4 $f(x) = px^3 + qx^2 - 37x - 12q$ where p and q are constants.

When $f'(x)$ is divided by $(x + 2)$ the remainder is -33

Given that $(x + 5)$ is a factor of $f(x)$

(a) (i) show that $p = 2$

(ii) find the value of q

(6)

(b) Hence, use algebra to factorise $f(x)$ completely.

(3)

(c) Hence solve the equation $f(x) = 0$

(2)

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6. Question 4 · part b · 3 marks · Working and answer

Working and answer

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7. Question 6 · part d · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part d only.

6 $f(x) = x^3 + (p + 1)x^2 - 10x + q$

where p and q are integers.

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that $9p + q + 6 = 0$

(3)

Given that $(x + p)$, where $p > 0$, is also a factor of $f(x)$

(b) show that $p^2 + 10p + q = 0$

(3)

(c) Hence find the value of p and the value of q.

(5)

(d) Using your values of p and q, factorise $f(x)$ completely.

(2)

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7. Question 6 · part d · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part d only.

Question 6 continued

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7. Question 6 · part d · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part d only.

Question 6 continued

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7. Question 6 · part d · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part d only.



Question 6 continued

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(Total for Question 6 is 13 marks)

7. Question 6 · part d · 2 marks · Working and answer

Working and answer

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8. Question 9 · part a · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part a only.

9 $f(x) = 3x^4 + 4x^3 - 36x^2 + 64$

Given that $f(x)$ can be written in the form $(x-2)^2(ax^2 + bx + c)$

(a) find the value of a, the value of b and the value of c

(4)

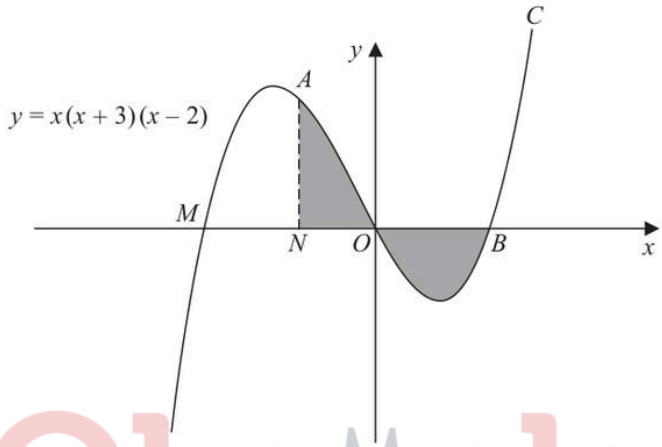


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a sketch of part of the curve C with equation $y = x(x+3)(x-2)$

The curve C crosses the x-axis at the point M, the origin and the point B.
The point N lies on the x-axis between M and O.
The point A lies on C such that AN is parallel to the y-axis.

The area of the shaded region bounded by the curve and OB is numerically equal to the area of the shaded region bounded by the curve, ON and NA.

Given that the coordinates of N are $(n, 0)$,

(b) use algebraic integration to show that n satisfies the equation

$$(x-2)^2(3x^2 + 16x + 16) = 0$$

(7)

(c) Hence find the exact coordinates of A.

(5)

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8. Question 9 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 5 · part b · 2 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

5 $f(x) = 2ax^3 + x^2 - bx + 3a$ where a and b are integers.

Given that $(x + 2)$ and $(x - 1)$ are both factors of $f(x)$

(a) show that $a = 2$ and find the value of b

(5)

(b) Hence factorise $f(x)$ completely.

(2)

Hence, given that $h(y) = 2^{(3y+2)} + 2^{2y} - 11(2^y) + 6$

(c) solve the equation $h(y) = 0$

Where appropriate give your answers to 3 decimal places.

(5)

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9. Question 5 · part b · 2 marks · Working and answer

Working and answer

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10. Question 5 · part c · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

5 $f(x) = 2x^3 + ax^2 - 14x + b$ where a and b are constants.

When $f(x)$ is divided by $(x - 4)$ the remainder is 39

Given that $(x - 1)$ is a factor of $f(x)$

(a) show that $a = -3$ and find the value of b

(5)

(b) Hence factorise $f(x)$ completely.

(4)

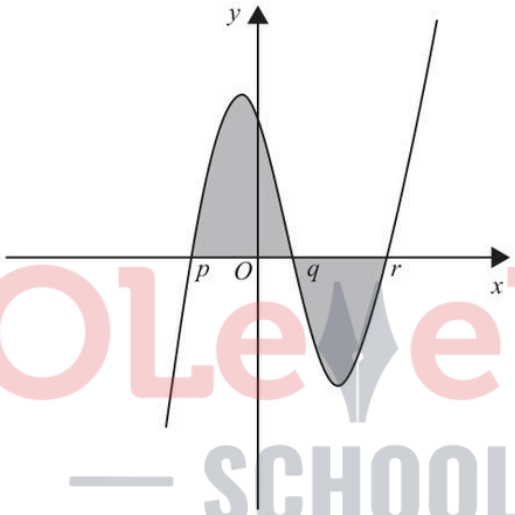


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = f(x)$

Given that C crosses the x-axis at the points with coordinates $(p, 0)$, $(q, 0)$ and $(r, 0)$

(c) write down the value of p, the value of q and the value of r

(3)

The region shown shaded in Figure 3 is bounded by the curve and the x-axis.

(d) Use algebraic integration to find the exact area of the shaded region.

(4)

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10. Question 5 · part c · 3 marks · Working and answer

Working and answer

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11. Question 5 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

5 $f(x) = ax^3 + 5bx^2 + 8ax - 4b$ where a and b are integers

Given that

$(x + 2)$ is a factor of $f(x)$

and that

when $f(x)$ is divided by $(x + 3)$ the remainder is 21

(a) show that $a = 2$ and find the value of b

(5)

(b) Use algebra to solve the equation $f(x) = 0$

(4)

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11. Question 5 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

Question 5 continued

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(Total for Question 5 is 9 marks)

11. Question 5 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 4 · part b · 3 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

4 $f(x) = x^3 + px^2 + qx + 6$ where p and q are constants.

Given that $(x - 1)$ is a factor of $f(x)$ and that when $f(x)$ is divided by $(x + 1)$ the remainder is 8

(a) (i) show that $p = -2$

(ii) find the value of q

(6)

(b) Hence, solve the equation $f(x) = 0$

(3)

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12. Question 4 · part b · 3 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 9 marks)

12. Question 4 · part b · 3 marks · Working and answer

Working and answer

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13. Question 2 · part d · 1 mark

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

2 $f(x) = x^3 + px + q$ where p and q are constants.

The remainder when $f(x)$ is divided by $(x - 1)$ is -12

The remainder when $f(x)$ is divided by $(x - 4)$ is 30

(a) Find the value of p and the value of q.

(6)

Using your values of p and q

(b) show that $f(3) = 0$

(1)

(c) Express $f(x)$ as a product of linear factors.

(3)

(d) Hence solve the equation $f(x) = 0$

(1)

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P 6 2 2 8 4 A 0 4 3 2

13. Question 2 · part d · 1 mark

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.



Question 2 continued

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(Total for Question 2 is 11 marks)

13. Question 2 · part d · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 10 · part b · 4 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

10 $f(x) = 32x^3 - 33x + 1$

(a) Show that $f(1) = 0$

(1)

(b) Hence using an algebraic method solve $f(x) = 0$

(4)

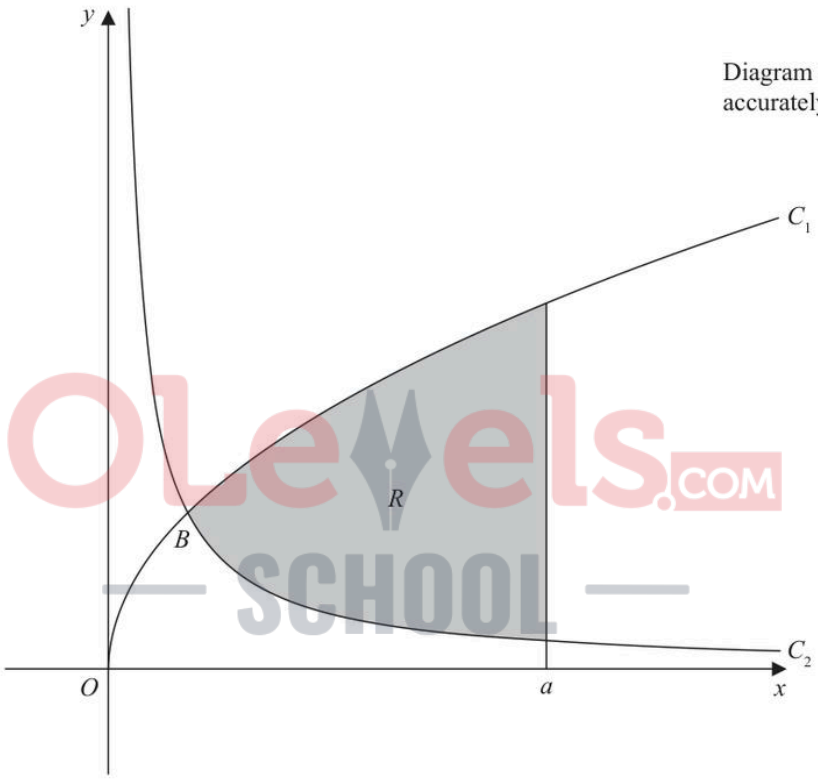


Diagram **NOT** accurately drawn

Figure 4

The region R, shown shaded in Figure 4, is bounded by the curve C_1 with equation $y = \sqrt{x}$, by the curve C_2 with equation $y = \frac{1}{8x}$ and by the line with equation $x = a$

The curves C_1 and C_2 intersect at the point B, with x coordinate p, where $p < a$

(c) Find the value of p.

(2)

The region R is rotated through 360° about the x-axis to generate a solid with volume $\frac{27\pi}{64}$

(d) Use algebraic integration to find the value of a.

(7)

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14. Question 10 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

15. Question 4 · part c · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

4 $f(x) = 2x^3 + ax^2 + bx + 18$ where a and b are constants

When $f'(x)$ is divided by $(x - 2)$ the remainder is 5

Given that $(x - 2)$ is a factor of $f(x)$

(a) find the value of a and the value of b.

(6)

(b) Express $f(x)$ as a product of linear factors.

(3)

(c) Hence use algebra to solve the equation $f(x) = 0$

(2)

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15. Question 4 · part c · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 11 marks)

15. Question 4 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

16. Question 4 · part c · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

4 $f(x) = px^3 + qx^2 - 37x - 12q$ where p and q are constants.

When $f'(x)$ is divided by $(x + 2)$ the remainder is -33

Given that $(x + 5)$ is a factor of $f(x)$

(a) (i) show that $p = 2$

(ii) find the value of q

(6)

(b) Hence, use algebra to factorise $f(x)$ completely.

(3)

(c) Hence solve the equation $f(x) = 0$

(2)

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16. Question 4 · part c · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

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Question 4 continued

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(Total for Question 4 is 11 marks)

16. Question 4 · part c · 2 marks · Working and answer

Working and answer

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17. Question 7 · part b · 3 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

7 $f(x) = 2x^3 + Ax^2 + Bx - 20$ where A and B are constants.

Given that

- $(2x+1)$ is a factor of $f(x)$
- when $f'(x)$ is divided by $(x-1)$ the remainder is -27

(a) show that $A = 3$ and find the value of B

(5)

(b) Hence, using algebra, solve the equation $f(x) = 0$

(3)

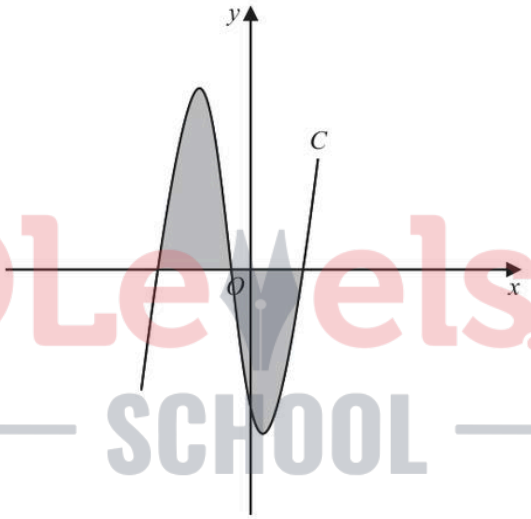


Figure 2

Figure 2 shows a sketch of part of the curve C with equation $y = f(x)$

(c) Use calculus to find the exact area of the finite region bounded by C and the x-axis, shown shaded in Figure 2

(4)

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17. Question 7 · part b · 3 marks · Working and answer

Working and answer

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18. Question 11 · part b · 6 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

11 $f'(x) = ax^2 - 14x - 10$ where $a \in \mathbb{Z}$

Given that $(x - 4)$ is a factor of $f(x)$ and that when $f(x)$ is divided by $(x + 1)$ the remainder is 25

(a) show that $a = 6$

(6)

(b) Hence use algebra to solve the equation $f(x) = 0$

(6)

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18. Question 11 · part b · 6 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

Question 11 continued

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18. Question 11 · part b · 6 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

Question 11 continued

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(Total for Question 11 is 12 marks)

18. Question 11 · part b · 6 marks · Working and answer

Working and answer

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19. Question 7 · part b · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part b only.

7 (a) Use the factor theorem to show that $(4x-1)$ is a factor of

$$f(x) = 64x^3 - 64x^2 + 3$$

(2)

(b) Hence, or otherwise, find the exact roots of the equation

$$f(x) = 0$$

(4)

A geometric series G has first term a and common ratio r
The third term of G is 9 and the sum to infinity of G is 192

(c) Show that $64r^3 - 64r^2 + 3 = 0$

(3)

Given that r is a rational number

(d) write down the value of r

(1)

(e) show that $a = 144$

(2)

The sum to n terms of G is S_n

(f) Using logarithms, find the least value of n such that $S_n > 191.9$

(4)

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19. Question 7 · part b · 4 marks · Working and answer

Working and answer

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20. Question 3 · part b · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

3 $g'(x) = mx^2 - 10x - 37$ where m is an integer

The curve $y = g(x)$ passes through the point with coordinates $(1, 20)$

Given that $(x - 5)$ is a factor of $g(x)$

(a) show that $g(x) = 2x^3 - 5x^2 - 37x + 60$

(5)

(b) Hence, or otherwise, use algebra to solve the equation $g(x) = 0$

(3)

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20. Question 3 · part b · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 8 marks)

20. Question 3 · part b · 3 marks · Working and answer

Working and answer

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21. Question 8 · part b · 6 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

8

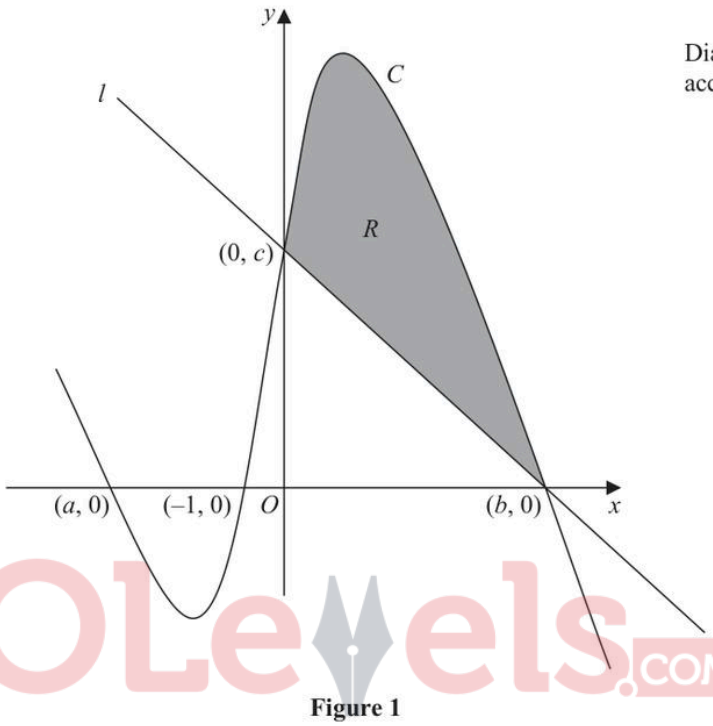


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows part of the curve C with equation $y = f(x)$

The curve C passes through the points with coordinates

$$(a, 0), (-1, 0), (b, 0) \text{ and } (0, c)$$

Given that $f'(x) = 17 + 2x - 3x^2$

(a) show that the equation of C is $y = 15 + 17x + x^2 - x^3$

(4)

(b) Find the value of a, the value of b and the value of c

(6)

The straight line l intersects C at the points with coordinates $(b, 0)$ and $(0, c)$

The region R, shown shaded in Figure 1, is bounded by l and C

(c) Use algebraic integration to find the exact area of region R

(5)

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21. Question 8 · part b · 6 marks · Working and answer

Working and answer

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22. Question 9 · part c · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.

9 $f(x) = 3x^4 + 4x^3 - 36x^2 + 64$

Given that $f(x)$ can be written in the form $(x-2)^2(ax^2 + bx + c)$

(a) find the value of a, the value of b and the value of c

(4)

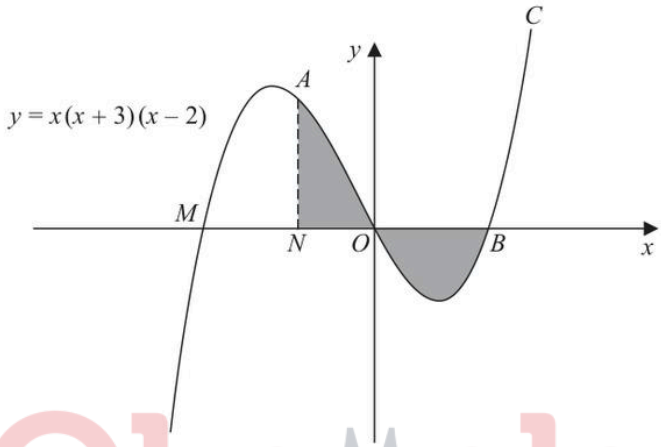


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a sketch of part of the curve C with equation $y = x(x+3)(x-2)$

The curve C crosses the x-axis at the point M, the origin and the point B.
The point N lies on the x-axis between M and O.
The point A lies on C such that AN is parallel to the y-axis.

The area of the shaded region bounded by the curve and OB is numerically equal to the area of the shaded region bounded by the curve, ON and NA.

Given that the coordinates of N are $(n, 0)$,

(b) use algebraic integration to show that n satisfies the equation

$$(x-2)^2(3x^2 + 16x + 16) = 0$$

(7)

(c) Hence find the exact coordinates of A.

(5)

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22. Question 9 · part c · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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22. Question 9 · part c · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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22. Question 9 · part c · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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(Total for Question 9 is 16 marks)

22. Question 9 · part c · 5 marks · Working and answer

Working and answer

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Practice workbook · Where a line meets a curve (simultaneous equations)

Solving one linear and one non-linear equation together, either stated as algebra or as a line and a curve (or two curves) on a diagram.

19 question parts: 1 worked-example retry and 18 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-13

1. Question 3 · part d · 4 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part d only.

3

$$f(x) = 8x^2 + 10x - 3$$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(3)

(b) Hence, or otherwise, find,

(i) the value of x for which $f(x)$ has a minimum,

(ii) the minimum value of $f(x)$.

(2)

The curve C has equation $y = f(x)$.

(c) Find the x coordinate of each of the points where C crosses the x-axis.

(2)

The straight line l has equation $y = 2x + 13$

(d) Use algebra to find the coordinates of the two points of intersection of C and l.

(4)

Using the same axes and the results of parts (b), (c) and (d),

(e) sketch the curve C and the straight line l.

(2)

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1. Question 3 · part d · 4 marks · Working and answer

Working and answer

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2. Question 5 · part whole · 6 marks

2019 JUN · Paper 02

Further practice · Attempt part whole only.

5 Use algebra to solve the equations

$$xy = 36$$

$$xy + x + 2y = 53$$

(6)

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2. Question 5 · part whole · 6 marks

2019 JUN · Paper 02

Further practice · Attempt part whole only.

Question 5 continued

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(Total for Question 5 is 6 marks)

2. Question 5 · part whole · 6 marks · Working and answer

Working and answer

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3. Question 2 · part a · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

2 One solution to the following simultaneous equations

$$y = px + 9$$
$$6x^2 - xy = 5$$

is $\left(-\frac{1}{2}, q\right)$, where p is an integer and q is a prime number.

(a) Find the value of p and the value of q

(4)

(b) Hence find the other solution to the equations.

(4)

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3. Question 2 · part a · 4 marks · Working and answer

Working and answer

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4. Question 2 · part b · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

2 One solution to the following simultaneous equations

$$y = px + 9$$
$$6x^2 - xy = 5$$

is $\left(-\frac{1}{2}, q\right)$, where p is an integer and q is a prime number.

(a) Find the value of p and the value of q

(4)

(b) Hence find the other solution to the equations.

(4)

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4. Question 2 · part b · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 8 marks)

4. Question 2 · part b · 4 marks · Working and answer

Working and answer

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5. Question 6 · part c · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

6 $f(x) = x^3 + (p + 1)x^2 - 10x + q$

where p and q are integers.

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that $9p + q + 6 = 0$

(3)

Given that $(x + p)$, where $p > 0$, is also a factor of $f(x)$

(b) show that $p^2 + 10p + q = 0$

(3)

(c) Hence find the value of p and the value of q.

(5)

(d) Using your values of p and q, factorise $f(x)$ completely.

(2)

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16

P 6 6 0 2 5 A 0 1 6 3 6

5. Question 6 · part c · 5 marks · Working and answer

Working and answer

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6. Question 2 · part c · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.

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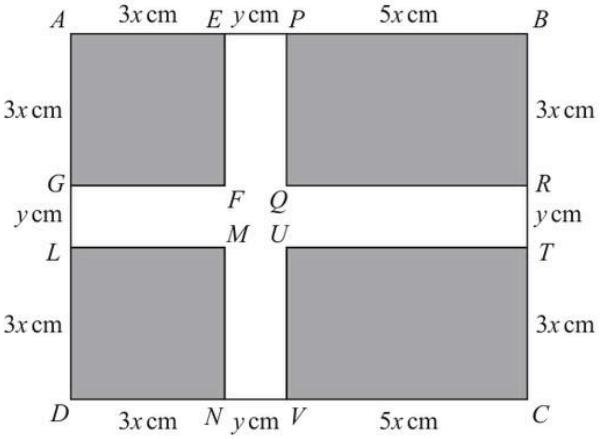


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the design for a flag consisting of a white cross on a grey background.

AEFG and *DLMN* are squares with sides of length $3x$ cm.
BPQR and *CTUV* are rectangles with sides of length $5x$ cm and $3x$ cm.
The width of the cross is y cm.

The total area of the flag is H cm^2

(a) Write down an expression, in terms of x and y, for H.

(1)

Given that the area of the cross is K cm^2

(b) show that $K = 14xy + y^2$

(3)

The total area of the flag is to be 3432 cm^2 and the area of the cross is to be 1080 cm^2

(c) Find the value of x and the value of y

(5)

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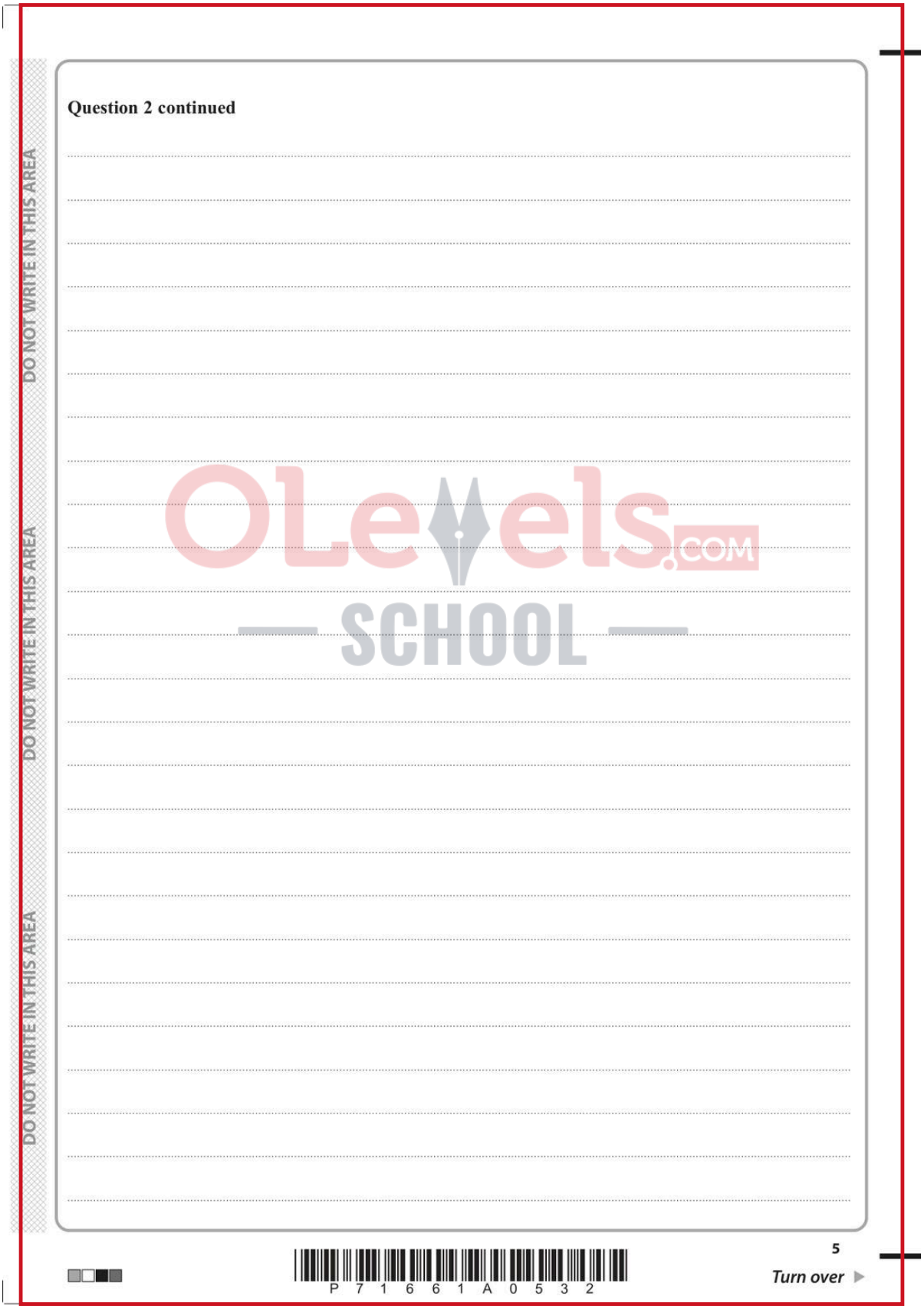
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6. Question 2 · part c · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.



Question 2 continued

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5

Turn over

6. Question 2 · part c · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.

Question 2 continued

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6. Question 2 · part c · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.

Question 2 continued

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(Total for Question 2 is 9 marks)

6. Question 2 · part c · 5 marks · Working and answer

Working and answer

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7. Question 2 · part b · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

2 (a) Express $x^2 + 4x - 8$ in the form $(x + a)^2 + b$ where a and b are constants whose values are to be found.

(2)

(b) Use algebra to solve the simultaneous equations

$$y = x^2 + 4x - 8$$
$$y = 2x + 7$$

(5)

The curve C has equation $y = x^2 + 4x - 8$
The straight line L has equation $y = 2x + 7$

Using the same axes and the results of part (a) and part (b),

(c) sketch C and L, showing clearly the coordinates of the turning point of C and the coordinates of the points of intersection of C and L.

(4)

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7. Question 2 · part b · 5 marks · Working and answer

Working and answer

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8. Question 10 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

10 $f(x) = 6x - x^2 \qquad x \in \mathbb{R}$

Given that $f(x)$ can be written in the form $D(x + E)^2 + F$ where D, E and F are integers,

(a) find the value of D, the value of E and the value of F.

(3)

(b) Find

(i) the maximum value of $f(x)$,

(ii) the value of x for which the maximum occurs.

(2)

The curve C has equation $y = f(x)$

The curve S has equation $y = x^2 - 4x + 8$

The curve S intersects the curve C at two points.

(c) Find the coordinates of each of these two points.

(4)

The finite region R is bounded by the curve C and the curve S.

(d) Use algebraic integration to find the area of R.

(4)

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8. Question 10 · part c · 4 marks · Working and answer

Working and answer

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9. Question 7 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

7

$$f(x) = x^2 - 9x + 14$$

Given that $f(x)$ can be written in the form $(x + a)^2 + b$, where a and b are constants,

(a) find the value of a and the value of b.

(2)

(b) Hence, or otherwise, find

(i) the minimum value of $f(x)$

(ii) the value of x for which this minimum occurs.

(2)

The curve C has equation $y = f(x)$

The line l has equation $y = x + 5$

(c) Use algebra to find the coordinates of the points of intersection of C and l.

(4)

(d) Use algebraic integration to find the exact area of the finite region bounded by C and l.

(5)

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9. Question 7 · part c · 4 marks · Working and answer

Working and answer

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10. Question 5 · part b · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

5 Given that k is a non-zero constant

curve C has equation $kx^2 - xy + (k+1)x = 1$

straight line l has equation $y = \frac{k}{2}x + 1$

The point A is the only point that lies on both C and l.

(a) Find the value of k

(6)

(b) Hence, find the coordinates of A.

(2)

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10. Question 5 · part b · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

Question 5 continued

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(Total for Question 5 is 8 marks)

10. Question 5 · part b · 2 marks · Working and answer

Working and answer

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11. Question 10 · part a · 4 marks

2023 JAN · Paper 01

Further practice · Attempt part a only.

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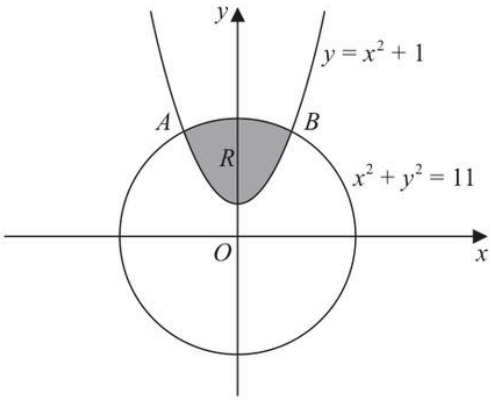


Diagram **NOT** accurately drawn

Figure 2

The region R, shown shaded in Figure 2, is bounded by the curve with equation $y = x^2 + 1$ and the curve with equation $x^2 + y^2 = 11$

The two curves intersect at the point A and at the point B.

(a) Find the x coordinate of the point A and the x coordinate of the point B.

(4)

The region R is rotated through 360° about the x-axis.

(b) Use algebraic integration to find the volume, to 2 decimal places, of the solid generated.

(5)

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11. Question 10 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 10 · part c · 2 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part c only.

10 $f(x) = 32x^3 - 33x + 1$

(a) Show that $f(1) = 0$

(1)

(b) Hence using an algebraic method solve $f(x) = 0$

(4)

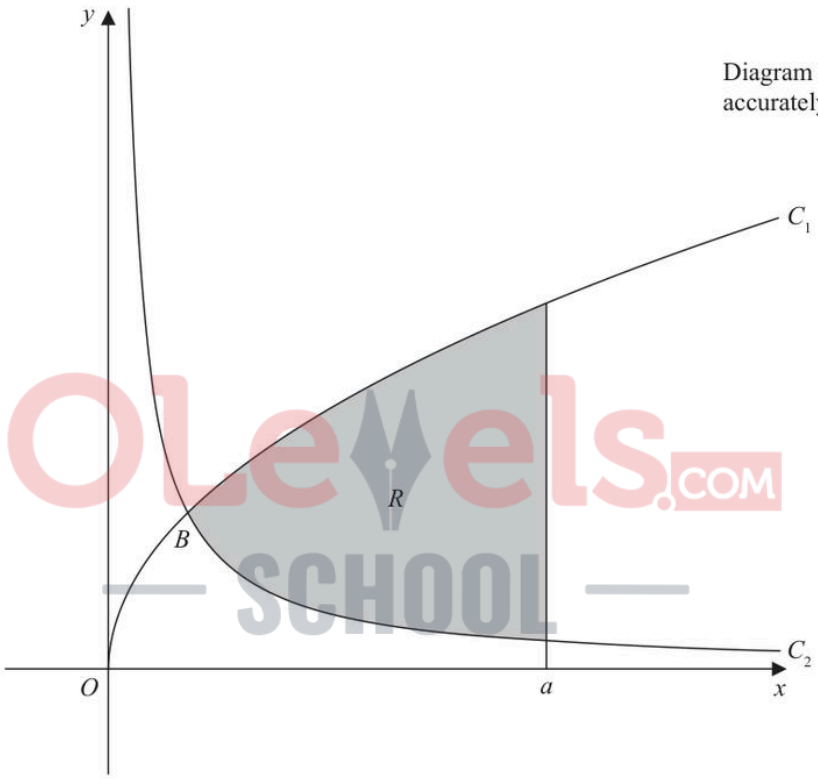


Diagram **NOT** accurately drawn

Figure 4

The region R, shown shaded in Figure 4, is bounded by the curve C_1 with equation $y = \sqrt{x}$, by the curve C_2 with equation $y = \frac{1}{8x}$ and by the line with equation $x = a$

The curves C_1 and C_2 intersect at the point B, with x coordinate p, where $p < a$

(c) Find the value of p.

(2)

The region R is rotated through 360° about the x-axis to generate a solid with volume $\frac{27\pi}{64}$

(d) Use algebraic integration to find the value of a.

(7)

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12. Question 10 · part c · 2 marks · Working and answer

Working and answer

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13. Question 5 · part a · 4 marks

2020 JAN · Paper 01

Further practice · Attempt part a only.

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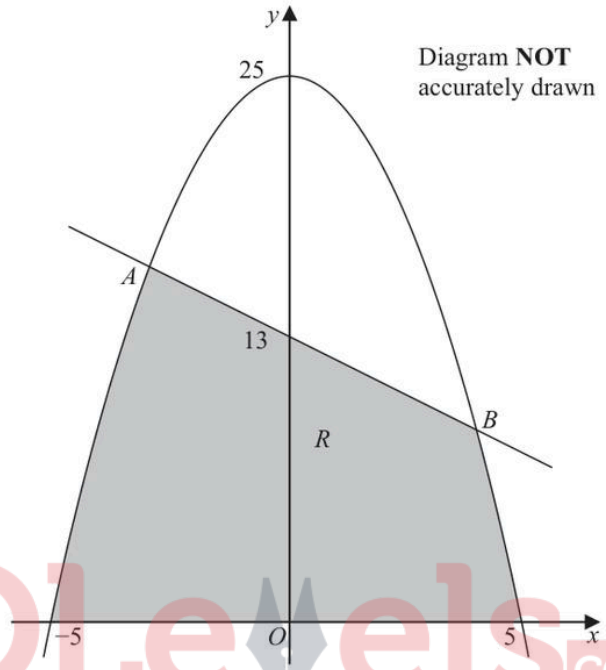


Figure 2

Figure 2 shows part of the curve with equation $y = 25 - x^2$ and part of the line with equation $y + x = 13$

The curve and the line intersect at the points A and B.

(a) Use algebra to find the coordinates of A and the coordinates of B.

(4)

The region R, shown shaded in Figure 2, is bounded by the curve, the straight line and the x-axis.

(b) Use algebraic integration to find the area of R.

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13. Question 5 · part a · 4 marks · Working and answer

Working and answer

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14. Question 11 · part c · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

11 $f(x) = 10 + 6x - x^2$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence, or otherwise, find

(i) the value of x for which $f(x)$ has its greatest value

(ii) the greatest value of $f(x)$

(2)

The curve C has equation $y = f(x)$

The curve S with equation $y = x^2 - x + 13$ intersects curve C at two points.

(c) Find the x coordinate of each of these two points.

(3)

(d) Use algebraic integration to find the exact area of the finite region bounded by the curve C and the curve S.

(5)

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14. Question 11 · part c · 3 marks · Working and answer

Working and answer

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15. Question 6 · part a · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

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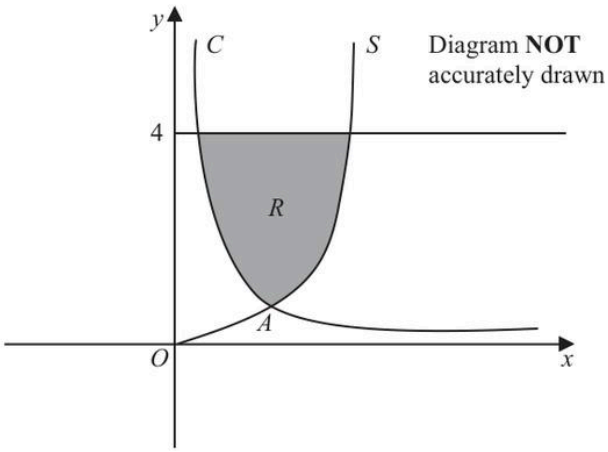


Figure 3

Figure 3 shows part of the curve C with equation $y = \frac{1}{4x}$, $x > 0$ and part of the curve S with equation $y = 2x^2$, $x \geqslant 0$

The curve C and the curve S intersect at the point A

(a) Find the coordinates of point A

(3)

The finite region R, shown shaded in Figure 3, bounded by the curve C, the curve S and the straight line $y = 4$ is rotated through 360° about the y-axis.

(b) Find, using algebraic integration, the exact volume of the solid formed.

(7)

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15. Question 6 · part a · 3 marks · Working and answer

Working and answer

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16. Question 2 · part a · 3 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

2 (a) Use algebra to find the x coordinates of the points where the curve with equation $y = 3x^2 + 9x - 17$ intersects the line with equation $y = 3 - 2x$

(3)

(b) Hence, or otherwise, find the set of values of x for which $3 - 2x \leqslant 3x^2 + 9x - 17$

(2)

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16. Question 2 · part a · 3 marks · Working and answer

Working and answer

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17. Question 6 · part a · 2 marks

2020 JAN · Paper 02

Further practice · Attempt part a only.

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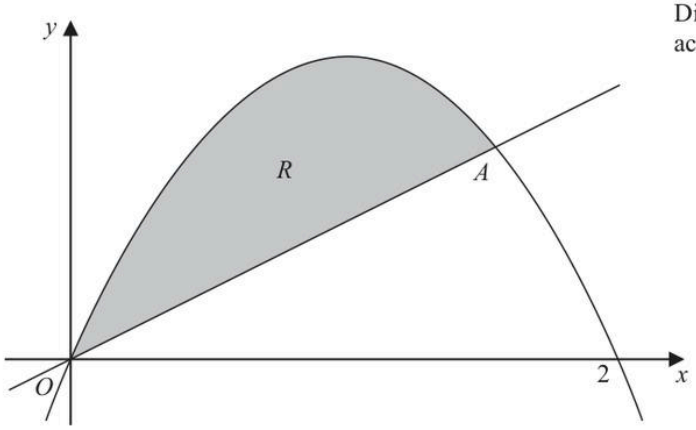


Diagram **NOT** accurately drawn

Figure 2

The region R, shown shaded in Figure 2, is bounded by the curve with equation $y = 2x - x^2$ and the line with equation $2y - x = 0$

The curve and the line intersect at the origin O and the point A.

(a) Show that the point A has coordinates $\left(\frac{3}{2}, \frac{3}{4}\right)$.

(2)

The region R is rotated through 360° about the x-axis.

(b) Use algebraic integration to find, in terms of π, the volume of the solid formed.

(6)

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17. Question 6 · part a · 2 marks · Working and answer

Working and answer

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18. Question 4 · part a.i · 0 marks

2023 JUN · Paper 02

Further practice · Attempt part a.i only.

4

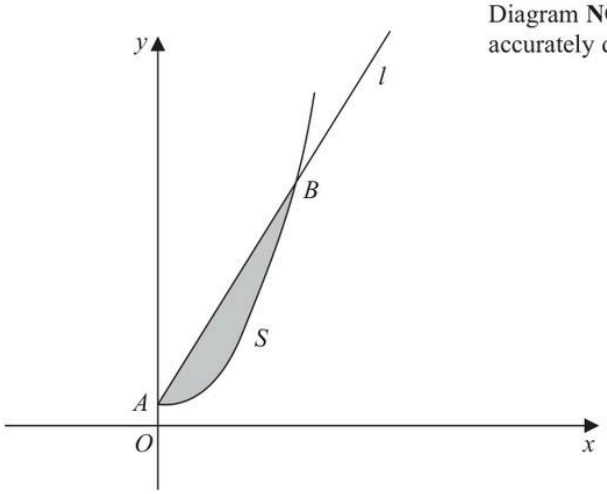


Diagram **NOT** accurately drawn

Figure 2

The curve S with equation $y = \frac{x^2}{4} + 2$ where $x \geqslant 0$ and the line l with equation $2y - x - 4 = 0$ where $x \geqslant 0$ intersect at the points A and B, as shown in Figure 2.

(a) (i) Show that the coordinates of point A are $(0, 2)$

(ii) Find the coordinates of the point B

(4)

The finite region bounded by S and l, shown shaded in Figure 2, is rotated through 2π radians about the y-axis.

(b) Use algebraic integration to find the volume of the solid generated. Give your answer in terms of π

(4)

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18. Question 4 · part a.i · 0 marks · Working and answer

Working and answer

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19. Question 4 · part a.ii · 4 marks

2023 JUN · Paper 02

Further practice · Attempt part a.ii only.

4

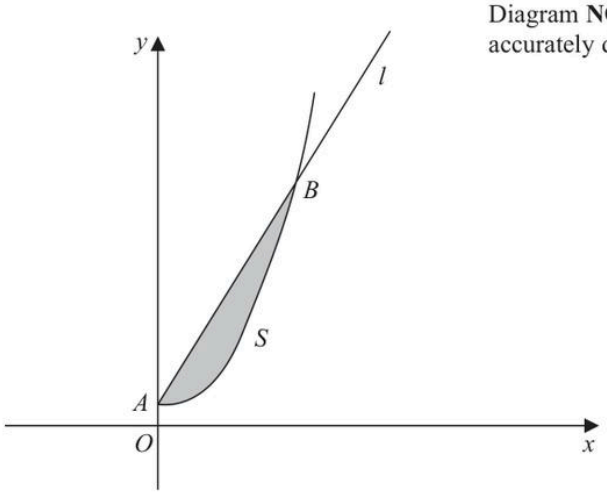


Diagram **NOT** accurately drawn

Figure 2

The curve S with equation $y = \frac{x^2}{4} + 2$ where $x \geqslant 0$ and the line l with equation $2y - x - 4 = 0$ where $x \geqslant 0$ intersect at the points A and B, as shown in Figure 2.

(a) (i) Show that the coordinates of point A are $(0, 2)$

(ii) Find the coordinates of the point B

(4)

The finite region bounded by S and l, shown shaded in Figure 2, is rotated through 2π radians about the y-axis.

(b) Use algebraic integration to find the volume of the solid generated. Give your answer in terms of π

(4)

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19. Question 4 · part a.ii · 4 marks · Working and answer

Working and answer

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Practice workbook · Quadratic inequalities

Rearranging an inequality into the form ax^2 + bx + c < 0 (or > 0), finding the critical values and choosing the correct region.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-02

1. Question 1 · part b · 4 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part b only.

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1 Find the set of values of x for which

(a) $2(3x - 1) < 4 - 3x$

(2)

(b) $3x^2 - 8x - 3 < 0$

(4)

(c) **both** $2(3x - 1) < 4 - 3x$ **and** $3x^2 - 8x - 3 < 0$

(1)

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(Total for Question 1 is 7 marks)

1. Question 1 · part b · 4 marks · Working and answer

Working and answer

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2. Question 2 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

2 $f(x) = 2x^2 - 12x + 5$

Given that $f(x)$ can be written in the form $f(x) = a(x + b)^2 + c$ where a, b and c are integers,

(a) find the value of a, the value of b and the value of c

(3)

(b) Hence find the set of values of x for which $f(x) - 37 > 0$

(4)

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2. Question 2 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 7 marks)

2. Question 2 · part b · 4 marks · Working and answer

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3. Question 1 · part b · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

1 Find the set of values for x for which

(a) $5x - 10 > 4x - 7$

(1)

(b) $2x^2 - 11x + 5 < 0$

(3)

(c) **both** $5x - 10 > 4x - 7$ **and** $2x^2 - 11x + 5 < 0$

(1)

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(Total for Question 1 is 5 marks)

3. Question 1 · part b · 3 marks · Working and answer

Working and answer

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4. Question 2 · part b · 4 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

2 Find the set of values of x for which

(a) $3x - 8 < 5x + 3$

(1)

(b) $4x^2 - 7x + 1 > 6 - 2x^2$

(4)

(c) **both** $3x - 8 < 5x + 3$ **and** $4x^2 - 7x + 1 > 6 - 2x^2$

(1)

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4. Question 2 · part b · 4 marks · Working and answer

Working and answer

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5. Question 4 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

4

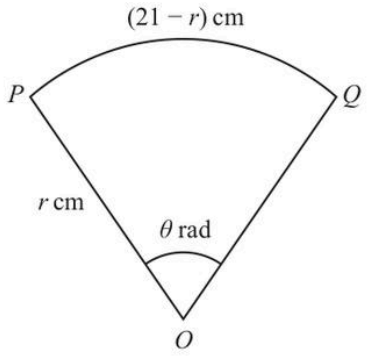


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the sector OPQ of a circle with centre O and radius r cm.

$OP = OQ = r$ cm arc $PQ = (21 - r)$ cm $\angle POQ = \theta$ radians

The area of the sector is A cm^2

(a) Show that $A = \frac{r}{2}(21 - r)$ **(3)**

The area of the sector must be greater than or equal to 27 cm^2

(b) Find the set of possible values of r **(4)**

(c) Hence write down the set of possible values of θ **(2)**

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5. Question 4 · part b · 4 marks · Working and answer

Working and answer

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6. Question 2 · part b · 3 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

2 Find the set of values for x for which

(a) $9-3x>11x+2$

(1)

(b) $10x^2+7x<12$

(3)

(c) **both** $9-3x>11x+2$ **and** $10x^2+7x<12$

(1)

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6. Question 2 · part b · 3 marks · Working and answer

Working and answer

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7. Question 1 · part b · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

1 Find the set of values for x for which

(a) $8x - 7 < 5x + 5$

(2)

(b) $2x^2 - 5x - 3 > 0$

(3)

(c) **both** $8x - 7 < 5x + 5$ **and** $2x^2 - 5x - 3 > 0$

(1)

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(Total for Question 1 is 6 marks)

7. Question 1 · part b · 3 marks · Working and answer

Working and answer

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8. Question 2 · part b · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

2 Find the set of values of x for which

(a) $2(x+1) < 5x - 2$

(2)

(b) $3x^2 - x \leqslant 10$

(3)

(c) **both** $2(x+1) < 5x - 2$ **and** $3x^2 - x \leqslant 10$

(1)

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8. Question 2 · part b · 3 marks · Working and answer

Working and answer

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9. Question 2 · part b · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

2 Find the set of values of x for which

(a) $3 + 2x \leqslant x + 2$

(1)

(b) $8x^2 + 10x < 3$

(4)

(c) **both** $3 + 2x \leqslant x + 2$ **and** $8x^2 + 10x < 3$

(1)

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9. Question 2 · part b · 4 marks · Working and answer

Working and answer

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10. Question 2 · part b · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

2 (a) Use algebra to find the x coordinates of the points where the curve with equation $y = 3x^2 + 9x - 17$ intersects the line with equation $y = 3 - 2x$

(3)

(b) Hence, or otherwise, find the set of values of x for which $3 - 2x \leqslant 3x^2 + 9x - 17$

(2)

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P 7 8 9 6 5 A 0 4 3 2

10. Question 2 · part b · 2 marks

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Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 5 marks)

10. Question 2 · part b · 2 marks · Working and answer

Working and answer

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Practice workbook · Two constants from factor and remainder information

Forming two equations in two unknown coefficients from factor and remainder information (sometimes about f'(x) or after integrating) and solving them simultaneously.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-09

1. Question 5 · part a · 5 marks

2022 JAN · Paper 01

Worked-example retry · Attempt part a only.

5 $f(x) = ax^3 + 5bx^2 + 8ax - 4b$ where a and b are integers

Given that

$(x + 2)$ is a factor of $f(x)$

and that

when $f(x)$ is divided by $(x + 3)$ the remainder is 21

(a) show that $a = 2$ and find the value of b

(5)

(b) Use algebra to solve the equation $f(x) = 0$

(4)

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1. Question 5 · part a · 5 marks · Working and answer

Working and answer

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2. Question 4 · part a.i · 0 marks

2023 JAN · Paper 01

Further practice · Attempt part a.i only.

4 $f(x) = x^3 + px^2 + qx + 6$ where p and q are constants.

Given that $(x - 1)$ is a factor of $f(x)$ and that when $f(x)$ is divided by $(x + 1)$ the remainder is 8

(a) (i) show that $p = -2$

(ii) find the value of q

(6)

(b) Hence, solve the equation $f(x) = 0$

(3)

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2. Question 4 · part a.i · 0 marks · Working and answer

Working and answer

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3. Question 4 · part a.ii · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part a.ii only.

4 $f(x) = x^3 + px^2 + qx + 6$ where p and q are constants.

Given that $(x - 1)$ is a factor of $f(x)$ and that when $f(x)$ is divided by $(x + 1)$ the remainder is 8

(a) (i) show that $p = -2$

(ii) find the value of q

(6)

(b) Hence, solve the equation $f(x) = 0$

(3)

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3. Question 4 · part a.ii · 6 marks · Working and answer

Working and answer

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4. Question 2 · part a · 6 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part a only.

2 $f(x) = x^3 + px + q$ where p and q are constants.

The remainder when $f(x)$ is divided by $(x - 1)$ is -12

The remainder when $f(x)$ is divided by $(x - 4)$ is 30

(a) Find the value of p and the value of q.

(6)

Using your values of p and q

(b) show that $f(3) = 0$

(1)

(c) Express $f(x)$ as a product of linear factors.

(3)

(d) Hence solve the equation $f(x) = 0$

(1)

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4. Question 2 · part a · 6 marks · Working and answer

Working and answer

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5. Question 5 · part a · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

5 $f(x) = x^3 + x^2 + x + c$ where c is a constant

The remainder when $f(x)$ is divided by $(x-2)$ is 4 times the remainder when $f(x)$ is divided by $(x+1)$

(a) Show that $c = 6$

(4)

Given that $(x+2)$ is a factor of $f(x)$

(b) show that the equation $f(x) = 0$ has only one real root.

(4)

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5. Question 5 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 4 · part a · 6 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

4 $f(x) = 2x^3 + ax^2 + bx + 18$ where a and b are constants

When $f'(x)$ is divided by $(x - 2)$ the remainder is 5

Given that $(x - 2)$ is a factor of $f(x)$

(a) find the value of a and the value of b.

(6)

(b) Express $f(x)$ as a product of linear factors.

(3)

(c) Hence use algebra to solve the equation $f(x) = 0$

(2)

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6. Question 4 · part a · 6 marks · Working and answer

Working and answer

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7. Question 5 · part a · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

5 $f(x) = 2x^3 + ax^2 - 14x + b$ where a and b are constants.

When $f(x)$ is divided by $(x - 4)$ the remainder is 39

Given that $(x - 1)$ is a factor of $f(x)$

(a) show that $a = -3$ and find the value of b

(5)

(b) Hence factorise $f(x)$ completely.

(4)

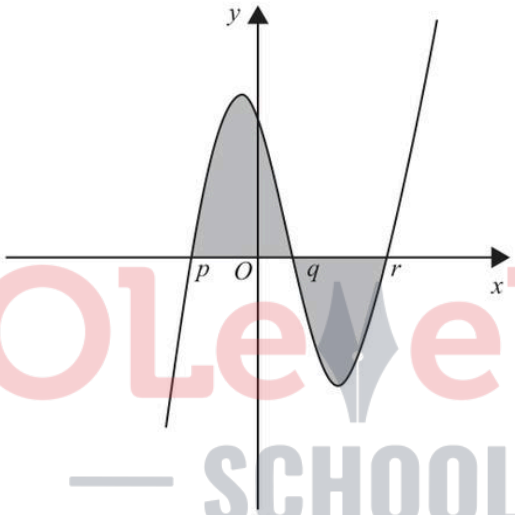


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = f(x)$

Given that C crosses the x-axis at the points with coordinates $(p, 0)$, $(q, 0)$ and $(r, 0)$

(c) write down the value of p, the value of q and the value of r

(3)

The region shown shaded in Figure 3 is bounded by the curve and the x-axis.

(d) Use algebraic integration to find the exact area of the shaded region.

(4)

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P 7 4 0 9 8 A 0 1 2 3 2

7. Question 5 · part a · 5 marks · Working and answer

Working and answer

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8. Question 4 · part a.i · 0 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

4 $f(x) = px^3 + qx^2 - 37x - 12q$ where p and q are constants.

When $f'(x)$ is divided by $(x + 2)$ the remainder is -33

Given that $(x + 5)$ is a factor of $f(x)$

(a) (i) show that $p = 2$

(ii) find the value of q

(6)

(b) Hence, use algebra to factorise $f(x)$ completely.

(3)

(c) Hence solve the equation $f(x) = 0$

(2)

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8. Question 4 · part a.i · 0 marks · Working and answer

Working and answer

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9. Question 4 · part a.ii · 6 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

4 $f(x) = px^3 + qx^2 - 37x - 12q$ where p and q are constants.

When $f'(x)$ is divided by $(x + 2)$ the remainder is -33

Given that $(x + 5)$ is a factor of $f(x)$

(a) (i) show that $p = 2$

(ii) find the value of q

(6)

(b) Hence, use algebra to factorise $f(x)$ completely.

(3)

(c) Hence solve the equation $f(x) = 0$

(2)

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9. Question 4 · part a.ii · 6 marks · Working and answer

Working and answer

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10. Question 7 · part a · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

7 $f(x) = 2x^3 + Ax^2 + Bx - 20$ where A and B are constants.

Given that

- $(2x+1)$ is a factor of $f(x)$
- when $f'(x)$ is divided by $(x-1)$ the remainder is -27

(a) show that $A = 3$ and find the value of B

(5)

(b) Hence, using algebra, solve the equation $f(x) = 0$

(3)

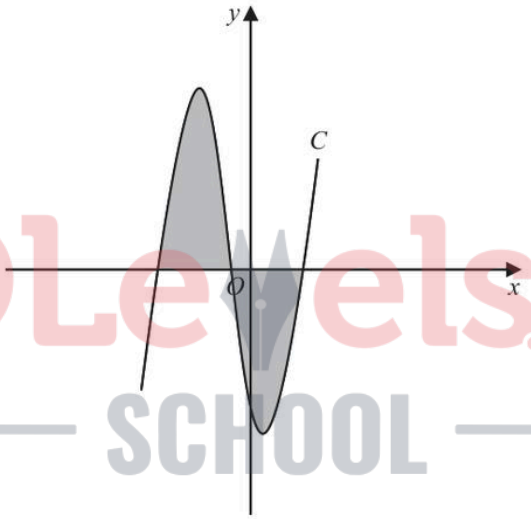


Figure 2

Figure 2 shows a sketch of part of the curve C with equation $y = f(x)$

(c) Use calculus to find the exact area of the finite region bounded by C and the x-axis, shown shaded in Figure 2

(4)

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P 7 8 9 6 5 A 0 1 4 3 2

10. Question 7 · part a · 5 marks · Working and answer

Working and answer

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11. Question 11 · part a · 6 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

11 $f'(x) = ax^2 - 14x - 10$ where $a \in \mathbb{Z}$

Given that $(x - 4)$ is a factor of $f(x)$ and that when $f(x)$ is divided by $(x + 1)$ the remainder is 25

(a) show that $a = 6$

(6)

(b) Hence use algebra to solve the equation $f(x) = 0$

(6)

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11. Question 11 · part a · 6 marks · Working and answer

Working and answer

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12. Question 4 · part a · 5 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

4 $f(x) = x^3 + px^2 + qx + 7$ where p and q are integers.

$(x + 1)$ is a factor of $f(x)$
The remainder when $f(x)$ is divided by $(x + 2)$ is -5

(a) Find the value of p and the value of q

(5)

(b) Hence, show that $f(x) = 0$ has only one real root.

(3)

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12. Question 4 · part a · 5 marks · Working and answer

Working and answer

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13. Question 5 · part a · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

5 $f(x) = 2ax^3 + x^2 - bx + 3a$ where a and b are integers.

Given that $(x + 2)$ and $(x - 1)$ are both factors of $f(x)$

(a) show that $a = 2$ and find the value of b

(5)

(b) Hence factorise $f(x)$ completely.

(2)

Hence, given that $h(y) = 2^{(3y+2)} + 2^{2y} - 11(2^y) + 6$

(c) solve the equation $h(y) = 0$

Where appropriate give your answers to 3 decimal places.

(5)

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13. Question 5 · part a · 5 marks · Working and answer

Working and answer

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